

Technical Handbook

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System	Infrastructure Webshop
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Part I

Overview & Architecture

Chapter 1

Introduction

1.1 Purpose of This Handbook

Open Hybrid Cloud is an open-source, self-service portal that enables Admins and Project Managers to order, manage, and decommission IT infrastructure on Linode/Akamai Cloud through a browser-based interface. The source code is publicly available at <https://github.com/Maximilian-Maag/open-hybrid-cloud>. This handbook is the authoritative reference for everyone who operates, extends, or uses the system.

It covers:

- The complete system architecture described with the C4 model
- Infrastructure-as-Code modules and the GitLab CI/CD concept
- Architecture and design decisions with their rationale
- Step-by-step administrator operations
- Developer onboarding, code structure, and extension patterns
- The full product lifecycle from creation to decommissioning

1.2 Target Audiences

Admin	Responsible for approving orders, monitoring all infrastructure, and ordering directly without an approval step. Reads Part III and Part V .
Root	Responsible for the product catalog, system configuration, and local user accounts. Reads Part III entirely.
Developer	Extends or maintains the webshop codebase and the OpenTofu modules. Reads Part I and Part IV .
Project Manager	Places infrastructure orders and manages projects. Reads Part V .

1.3 Conventions Used in This Handbook

monospace File paths, commands, code, environment variables, and route patterns.

italic Technical terms on first use.

bold UI labels, button names, and menu items.

Note boxes Yellow-highlighted boxes providing additional context or important background information.

Important boxes Orange-highlighted boxes warning about actions that can cause data loss or service disruption.

Tip boxes Green-highlighted boxes with shortcuts and best practices.

1.4 System At a Glance

The webshop is a single Go binary that renders HTML server-side and delivers page fragments via HTMX for a responsive, app-like feel. Orders are fulfilled by triggering GitLab CI/CD pipelines that run OpenTofu (a Terraform-compatible open-source tool) to provision or destroy cloud resources on Linode/Akamai. The system is stateless above the PostgreSQL layer, enabling horizontal scaling without coordination.

Table 1.1: Technology stack overview

Layer	Technology	Version
Backend	Go	1.22+
Templates	templ	0.3+
Frontend	HTMX	2.0
CSS	Tailwind CSS	3.x
Database	PostgreSQL	15+
DB Driver	pgx/v5 (jackc)	5.x
IaC	OpenTofu	1.9+
Cloud	Linode/Akamai	—
CI/CD	GitLab CE	16+
State Backend	GitLab Managed TF State	—
Auth	OIDC (Entra ID) + local	—

Chapter 2

C4 Architecture Model

The system is described using the *C4 model* (Context, Container, Component, Code). Each level zooms in one step deeper.

2.1 Level 1 — System Context

Figure 2.1 shows who uses the webshop and which external systems it depends on.

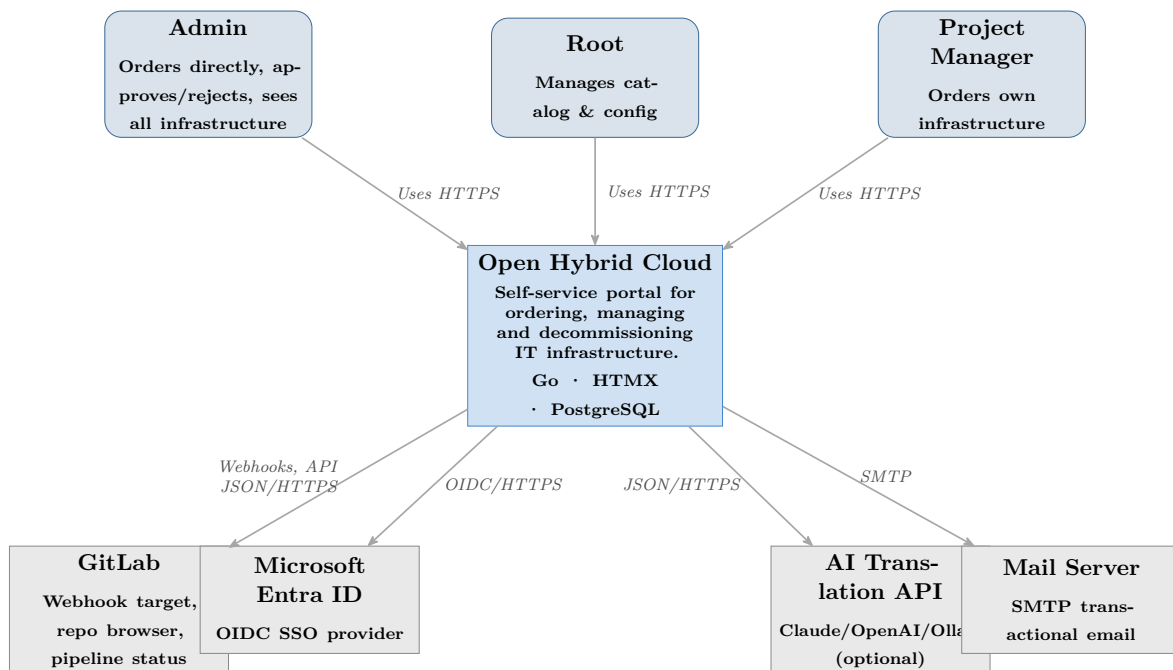


Figure 2.1: C4 Level 1 — System Context Diagram

2.1.1 Actors

Admin Authenticates via Microsoft Entra ID SSO. Has the system role `admin`. Can order infrastructure directly (no approval required), approve or reject orders from

Project Managers, view all projects and infrastructure, and decommission any resource.

Root Uses a local account (no SSO). Has the system role **root**. Manages the product catalog, system configuration (SMTP, AI, branding, currencies), environments, GitLab sources, cost centers, and local users. Cannot place orders.

Project Manager Authenticates via Entra ID SSO. Has the system role **user**. Places orders (which require Admin approval), manages their own projects, and can decommission their own infrastructure.

2.1.2 External Systems

GitLab One or more self-hosted or cloud GitLab instances. The webshop connects to GitLab for three purposes: (1) browsing repositories and importing `variables.tf` parameter definitions, (2) triggering provisioning and decommissioning webhooks, and (3) polling pipeline status to update order state.

Microsoft Entra ID The OIDC identity provider for Admins and Project Managers. The webshop implements the Authorization Code Flow. Local accounts for Root users bypass this provider entirely.

AI Translation API Optional. Used to auto-translate product names and descriptions into all 25 supported languages. Multiple backends are supported: cloud-based (Claude, OpenAI, Azure OpenAI) and on-premise (Ollama, LocalAI). The feature is hidden when no provider is configured.

Mail Server Any SMTP server. Sends order confirmations, approval request notifications, approval/rejection notices, deployment completion alerts, and failure notifications.

2.2 Level 2 — Container Diagram

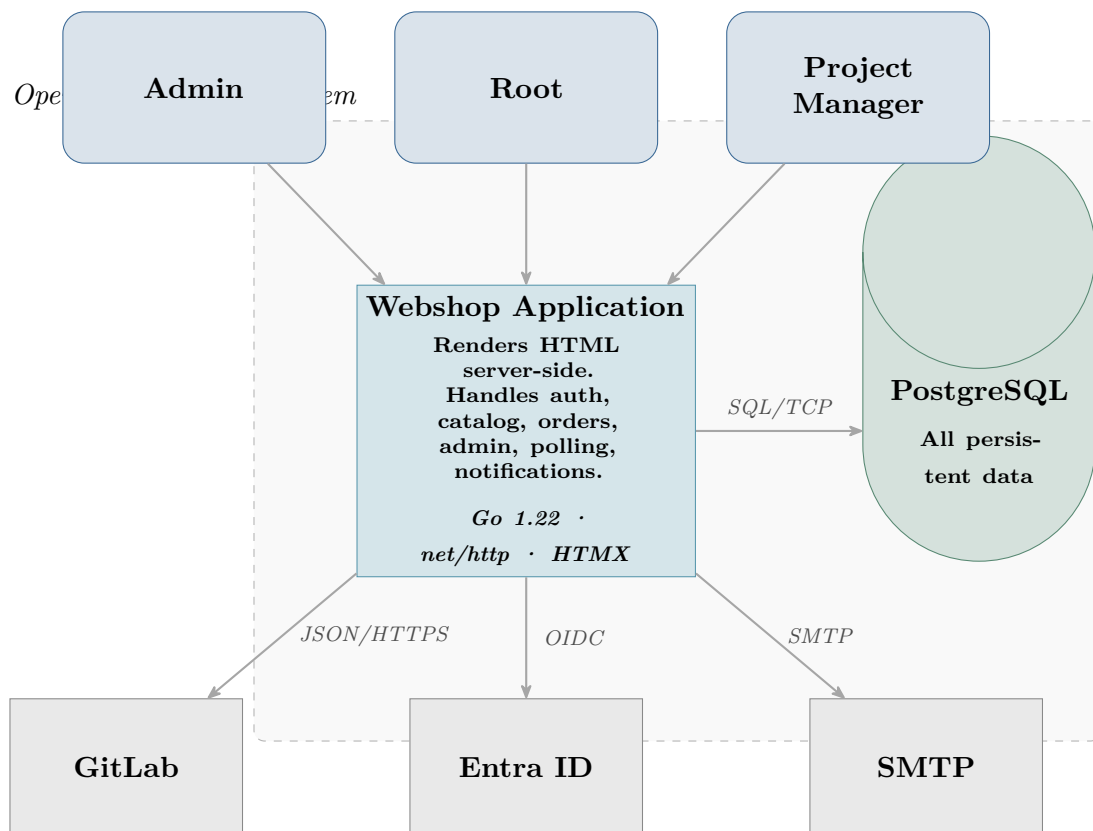


Figure 2.2: C4 Level 2 — Container Diagram

The system has exactly two containers:

Webshop Application A single stateless Go binary. It handles HTTP requests, renders HTML via `templ` components, serves HTMX fragments, coordinates all business logic, runs the GitLab polling goroutines, and manages outbound SMTP. Being stateless above the database layer means multiple replicas can run behind a load balancer without any coordination layer.

PostgreSQL Database The single source of truth. Stores the product catalog, orders, infrastructure state, audit log, branding, configuration, users, and exchange rates. The database is the only stateful component.

2.3 Level 3 — Component Diagram

Figure 2.3 decomposes the Go application container into its major logical components.

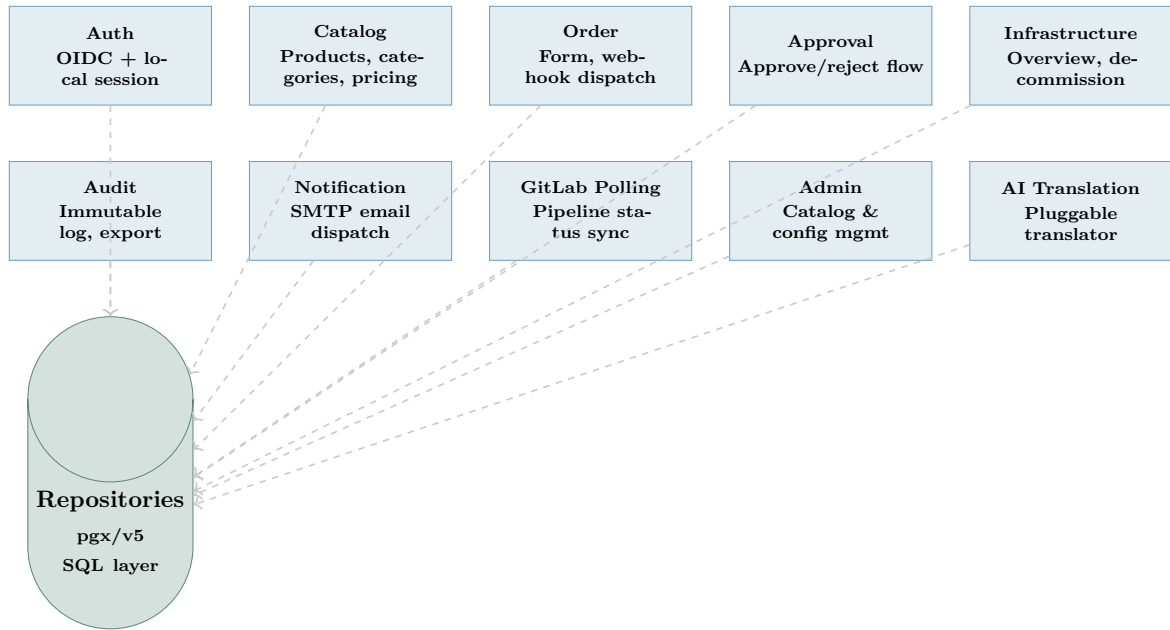


Figure 2.3: C4 Level 3 — Component Diagram (Webshop Application)

2.3.1 Component Descriptions

Component	Responsibility
Auth	Handles OIDC Authorization Code Flow (Entra ID) and local password login. Issues encrypted, HttpOnly session cookies. Enforces role-based access (<code>admin</code> , <code>root</code> , <code>project_leader</code>).
Catalog	Renders products grouped by category. Converts prices from the configured base currency to the user's locale currency using stored exchange rates. Supports 25 languages via the <code>i18n</code> engine.
Order	Renders the order form with parameters from global, category, and product parameter sets. Stores orders in PostgreSQL. For Admins, triggers the GitLab provisioning webhook immediately; for Project Managers, parks the order for approval.
Approval	Lists pending orders for Admins. Approve triggers the GitLab webhook. Reject requires a mandatory comment. Both paths send email notifications and write to the audit log.
Infrastructure	Shows deployed resources grouped by project and environment. Role-scoped: Project Managers see only their own. Triggers the GitLab destroy webhook for decommissioning. Supports “use as template” to pre-fill a new order.

Component	Responsibility
Audit	Append-only log of all system actions. Filterable by date range, user, and action type. Exportable as CSV or PDF.
Notification	Wraps SMTP email delivery. Silently no-ops when SMTP is not configured. SMTP credentials are live-reloadable from the database without a server restart.
GitLab Polling	Goroutine pool that periodically queries the GitLab Pipelines API for in-flight orders. Updates order and infrastructure state on completion or failure, then triggers email notifications. Uses PostgreSQL advisory locks so multiple replicas do not double-poll.
Admin	All Root pages: categories, products (CRUD, image upload, AI translation, webhook management, parameter import from <code>variables.tf</code>), environments, GitLab sources, cost centers, currencies, SMTP, AI config, branding, users.
AI Translation	Pluggable interface (<code>aitranslation.Translator</code>). Concrete implementations for Claude, OpenAI, Azure OpenAI, Ollama, and LocalAI. Selected and instantiated at startup from the database configuration; swappable at runtime via the Admin UI.
Repositories	Thin SQL layer using <code>pgx/v5</code> . One repository interface per aggregate root (User, Product, Order, etc.). Implementations in internal/repository/postgres/ .

Table 2.1: Component descriptions

2.4 Level 4 — Key Code Flows

2.4.1 Order Placement Flow (Project Manager)

1. Browser submits `POST /orders/new`.
2. `handler.orderCreate()` validates input, calls `service.OrderService.Create()`.
3. The order service writes the order record with status `pending_approval` to PostgreSQL.
4. The notification service sends an approval-request email to all Admins.
5. The audit service writes an `order_placed` entry.
6. Browser redirects to `GET /orders/{id}` showing the *Awaiting Approval* status.

2.4.2 Order Approval & Provisioning Flow

1. Admin opens `GET /approvals` and clicks **Approve**.
2. `handler.approvalApprove()` calls `service.OrderService.Approve()`.
3. The order service updates status to `provisioning`, stores the webhook URLs from the product-environment configuration.
4. For each configured webhook (ordered by `exec_order`): an HTTP POST is sent to the GitLab webhook URL with the Linode/Tofu parameters as JSON.
5. GitLab receives the trigger, starts a pipeline, returns a pipeline ID.
6. The webshop stores the pipeline ID(s) in the JSONB `pipeline_ids` column.
7. The GitLab Polling goroutine detects the in-flight pipeline and starts querying the Pipelines API every 30 seconds.
8. When the pipeline completes successfully, the poller sets status to `completed` and sends a deployment-complete email.
9. On failure, status becomes `failed` and failure emails go to the orderer and all Admins.

2.4.3 Request–Response Lifecycle (HTMX)

Most user interactions either perform a full page load (navigation) or return an HTML fragment (HTMX). The server detects HTMX requests via the `HX-Request` header. Full renders compose a templ `Layout` component around the page component. Fragments skip the layout and return only the relevant HTML snippet, which HTMX swaps into the DOM.

Chapter 3

Design Decisions & Technology Rationale

3.1 Go as Application Language

Decision: The entire backend is written in Go.

Rationale:

- Compiles to a single self-contained binary with no runtime dependencies — ideal for Docker-based deployment.
- Low memory footprint and predictable garbage collection enable running multiple polling goroutines without resource pressure.
- The standard library `net/http` in Go 1.22+ provides pattern-matching routing (`"GET /path/{id}"`) sufficient for this use case, removing the need for a router dependency.
- `html/template` provides automatic context-aware escaping, eliminating entire classes of XSS vulnerabilities.
- Goroutines provide a natural model for the concurrent GitLab pipeline polling without callback complexity.

Alternative considered: Python/FastAPI with Jinja2. Rejected because the deployment story is more complex (virtualenv, interpreter) and concurrency requires `async`, which increases cognitive overhead.

3.2 Server-Side Rendering with HTMX

Decision: Pages are rendered server-side; dynamic updates use HTMX HTML fragments rather than a JavaScript SPA framework.

Rationale:

- No JSON API surface to design, document, or secure separately — the server owns both data and presentation.
- Progressive enhancement: the UI degrades gracefully if JS is unavailable.
- HTMX’s `hx-get/hx-post` attributes on HTML elements cover the interactivity requirements (live status polling, inline form submission, partial page swaps) without React or Vue build pipelines.
- Template-level security: Go’s `html/template` auto-escapes all inserted values, preventing XSS by construction.

3.3 templ for Type-Safe HTML Components

Decision: All HTML pages and UI components are authored as `.templ` files and compiled to Go by the `templ` CLI. The `html/template` package is no longer used for page rendering.

Rationale:

- **Compile-time type safety:** Each page receives a concrete typed view struct (e.g. `view.OrderListView`). Passing the wrong data to a component is a compile error, not a runtime `nil` panic or a missing-key silently rendering empty.
- **No `map[string]any`:** The old `html/template` approach required every handler to pass `map[string]any{}` to `h.render()`, providing no compile-time guarantee that the template would find the keys it expected.
- **Composable components:** `templ` supports Go functions as components with typed parameters, children slots, and helper functions — the same mental model as React components but compiled to plain Go without a JS build step.
- **XSS safety preserved:** `templ` auto-escapes all string interpolations by default, maintaining the same security guarantee as `html/template`.
- **IDE support:** The `templ` VS Code extension provides syntax highlighting, autocompletion, and go-to-definition inside `.templ` files.

Design system: A shared component library under `ui/comp/` (`comp.Btn`, `comp.Card`, `comp.Badge`, `comp.Field`, etc.) provides a consistent visual language. Page components under `ui/pages/` compose these primitives and receive typed view models from the handler layer.

3.4 Tailwind CSS without DaisyUI

Decision: Pure Tailwind CSS utility classes; DaisyUI was removed after the migration to `templ` was complete.

Rationale:

- **Theme collision:** DaisyUI ships its own CSS custom properties (`-b1`, `-p`, `-s`, ...) which conflicted with the application's own branding variables (`-bp`, `-bs`). Keeping both required workarounds and caused unpredictable cascade ordering.
- **Specificity fights:** DaisyUI's component classes (`btn`, `badge`, `card`) carry higher specificity than plain utilities, making it hard to override individual properties without `!important` or extra wrapper classes.
- **Bundle size and opacity:** DaisyUI adds several hundred kilobytes of pre-defined component CSS. Tailwind's content-scan only purges *unused* utilities; DaisyUI component styles are included unconditionally, inflating the bundle with styles that the templ components never use.
- **Design independence:** The hand-authored component library in `ui/comp/` uses only Tailwind utilities, making every style decision explicit and auditable. There are no hidden DaisyUI defaults to debug.

What replaced it: The `ui/comp/` package provides the same functional coverage (buttons, badges, cards, form inputs, checkboxes, layout shell) using only Tailwind utilities and the two brand CSS variables `var(-bp)` and `var(-bs)`.

3.5 No Alpine.js

Decision: Alpine.js was removed from the project. The only remaining JavaScript beyond HTMX is the CDN-loaded Alpine.js on the branding admin page, where it powers the live colour preview.

Rationale:

- **All use cases replaced by HTML:** Every place Alpine.js was used — collapsible sections, toggle visibility, show/hide forms — can be handled by the native `<details>/<summary>` element. This is a zero-JavaScript, zero-dependency solution that is accessible by default and degrades gracefully in all browsers.
- **Reduced attack surface:** Alpine.js evaluates expressions from HTML attributes (`x-data`, `x-show`, `x-on:click`). Any XSS that manages to inject an attribute becomes executable code. Removing Alpine.js eliminates this vector.
- **Dependency hygiene:** Alpine.js loaded from a CDN introduces a third-party supply-chain dependency on every page load. Removing it means the only external script is HTMX, which is already trusted as a core dependency.
- **Consistency:** Mixing two reactive JavaScript libraries (HTMX and Alpine) creates a mental model split: developers must decide which tool owns each interactive behaviour. Pure HTMX + HTML is a single coherent model.

Exception: The admin branding page (`/admin/branding`) retains Alpine.js loaded via CDN script tag. It drives a live colour-preview widget where two colour inputs immediately update a mock header element without a server round-trip. This is the one genuine use case for in-page reactive state that cannot be reasonably replaced by `<details>` or HTMX alone.

3.6 PostgreSQL with pgx (No ORM)

Decision: PostgreSQL 15+, accessed via pgx/v5 with hand-written SQL. No ORM.

Rationale:

- JSONB arrays for `pipeline_ids` allow storing multiple concurrent pipeline references per order without a join table, while retaining full query ability.
- BYTEA columns store product images and the branding logo directly in the database, eliminating a separate object store for this scale.
- Direct SQL gives precise control over query shape and enables `ON CONFLICT DO UPDATE` (upserts) for the branding and app-config singleton rows.
- PostgreSQL advisory locks coordinate the polling goroutines across multiple replicas without an additional message broker.
- pgx is the highest-performance PostgreSQL driver for Go; it avoids the `database/sql` reflection overhead.

3.7 Single-Container Architecture

Decision: The entire application (HTTP server, background polling, notification dispatch) runs in a single container.

Rationale: At the anticipated load (tens of users, hundreds of orders per month), microservice decomposition would add operational complexity (service discovery, inter-service auth, distributed tracing) without meaningful benefit. The single-binary model keeps deployment simple: one `docker run` or one Kubernetes `Deployment` object. Horizontal scaling is achieved by running multiple replicas; PostgreSQL advisory locks ensure the polling goroutines do not duplicate work.

3.8 OpenTofu over Terraform

Decision: All IaC modules use OpenTofu 1.9+.

Rationale: OpenTofu is a community-governed, truly open-source fork of Terraform created after HashiCorp changed Terraform's license to BSL 1.1 in 2023. The API, HCL syntax, provider ecosystem, and state format are fully compatible. Using OpenTofu avoids future licensing costs and complies with Open Hybrid Cloud's open-source software policy.

3.9 GitLab Managed Terraform State

Decision: OpenTofu state is stored in GitLab's built-in HTTP-compatible state backend (GitLab Managed Terraform State).

Rationale:

- Reuses existing GitLab infrastructure — no S3 bucket, no Terraform Cloud account, no Vault setup required.
- State is protected by GitLab’s RBAC: only project members with Developer or higher access can read or modify state.
- State locking via HTTP LOCK/UNLOCK prevents concurrent apply operations.
- State is browsable in the GitLab UI under **Infrastructure** → **Terraform states**.

3.10 Per-Resource Isolated State

Decision: Each VM and each Kubernetes cluster has its own isolated OpenTofu state (named `vm-{label}` and `k8s-{label}`).

Rationale: Shared state across multiple VMs creates blast-radius risk: a failing `apply` for one VM could lock the state and block unrelated VMs. Isolated states mean each resource’s lifecycle is entirely independent. The trade-off is a larger number of state files, which is manageable at the expected scale.

3.11 Webhook-Driven Pipeline Trigger

Decision: The webshop triggers GitLab pipelines via webhook (HTTP POST) rather than the GitLab API’s pipeline-trigger endpoint.

Rationale: Webhooks are the native GitLab mechanism for external triggers. They carry a secure token for authentication and include the ability to pass variables (OpenTofu parameters) as JSON. The pipeline configuration (`.gitlab-ci.yml`) lives in the same repository as the IaC code, making the trigger configuration auditable and version-controlled.

3.12 Internationalization: 25 Languages

Decision: All UI strings are externalized in a key-value i18n map covering 25 languages (all EU official languages plus Russian and Ukrainian).

Rationale: Open Hybrid Cloud operates across Europe. The map is defined in `internal/i18n/i18n.go` as a Go map literal for compile-time completeness checking. Translation of product content (not UI strings) is delegated to the optional AI translation service.

3.13 Session Security: Encrypted Cookies

Decision: User sessions are stored in encrypted, signed HTTP cookies — not in the database or server-side session store.

Rationale:

- No database round-trip on every request to look up a session.
- Horizontal scaling is trivial: no shared session store required.
- The `SESSION_KEY` (32-byte, hex-encoded) is used to both sign and encrypt the cookie content via AES-GCM, so the server can verify integrity without a separate HMAC key.
- Cookies are set with `HttpOnly`, `Secure`, and `SameSite=Lax` to mitigate XSS session theft and CSRF.
- Logout invalidates the session by overwriting the cookie with an empty, expired value — no server-side revocation list needed for the MVP threat model.

Alternative considered: Server-side sessions in PostgreSQL. Rejected to avoid the extra table and join on every authenticated request. If revocation of individual sessions at scale is required this can be introduced as a future enhancement.

3.14 Password Hashing: bcrypt

Decision: Local user passwords are hashed with bcrypt (cost factor 12) via golang.org/x/crypto/bcrypt.

Rationale:

- bcrypt is an adaptive, memory-hard function specifically designed for password storage; it is resistant to GPU-accelerated dictionary attacks.
- Cost factor 12 takes 250ms on modern hardware — slow enough to deter brute-force while still acceptable for interactive login.
- The Go standard-library-adjacent `x/crypto` package is actively maintained by the Go team.
- Passwords are never stored in plaintext or with reversible encryption.

Alternative considered: Argon2id. More modern, but bcrypt is widely understood, well-tested in Go, and sufficient for the low login-volume use case.

3.15 Static Asset Embedding: embed.FS

Decision: HTML templates and static assets (CSS, JS, images) are embedded into the Go binary at compile time using the `//go:embed` directive and `embed.FS`.

Rationale:

- The resulting binary is fully self-contained — no external file system paths need to match the deployment environment.
- Deployment is a single `COPY` in the Dockerfile (multi-stage build); no volume mounts are required for templates.

- Template changes are compiled into the binary, so a deployment is always atomic: the binary and its templates are in sync.
- The Go compiler resolves embed paths at build time, making missing-file bugs a compile error rather than a runtime panic.

Trade-off: Hot-reloading templates during development is not possible without restarting the server. The `make dev` target runs the server with `go run`, so a simple restart is fast enough for the development workflow.

Part II

Infrastructure as Code

Chapter 4

GitLab Repository Concept

4.1 Overview

The IaC layer consists of six GitLab repositories in the `open-hybrid-cloud` group. Each repository is a self-contained OpenTofu module with its own CI/CD pipeline and state. Together they provision all cloud infrastructure on Linode/Akamai.

Table 4.1: GitLab repository overview

Repository	Purpose	Trigger
<code>network-core</code>	VPC, subnets, base firewall, DNS zone	Manual
<code>vm-platform</code>	Linode VM instances (per-VM)	Webhook
<code>k8s-platform</code>	LKE Kubernetes clusters (per-cluster)	Webhook
<code>shared-services</code>	GitLab CE + Vault + Registry	Manual
<code>observability</code>	Grafana + Prometheus + Loki	Manual
<code>infra-state</code>	State backend host (no IaC code)	N/A

4.2 Repository File Structure

Every module repository (except `infra-state`) follows the same file layout:

```
<module>/
+-- .gitlab-ci.yml # CI/CD pipeline definition
+-- backend.tf    # HTTP state backend (GitLab Managed)
+-- provider.tf   # Linode provider + required versions
+-- variables.tf  # Input variable declarations
+-- main.tf       # Resource definitions
+-- outputs.tf    # Output values
```

4.3 State Management

All state files live in the `infra-state` project. The project hosts no IaC code; it is solely a container for GitLab Managed Terraform State files.

State naming convention:

Table 4.2: State name convention per module

Module	State name
<code>network-core</code>	<code>network-core</code>
<code>vm-platform</code>	<code>vm-{vm_label}</code>
<code>k8s-platform</code>	<code>k8s-{cluster_label}</code>
<code>shared-services</code>	<code>shared-prod</code>
<code>observability</code>	<code>observability-prod</code>

The backend URL pattern is:

```
https://<gitlab-host>/api/v4/projects/<INFRA_STATE_PROJECT_ID>/terraform/state/<
  ↳ STATE_NAME>
```

Backend configuration is passed at `tofu init` time via `-backend-config` flags, not hard-coded in `backend.tf`. This keeps credentials out of version control.

4.4 CI/CD Pipeline Stages

All module pipelines share the same three-stage structure:

1. **validate** — `tofu validate`: checks HCL syntax and provider schema compliance.
2. **plan** — `tofu plan -out=tfplan`: computes the diff and saves the plan artifact.
3. **apply** — `tofu apply tfplan`: applies the plan. For webhook-triggered modules, apply runs automatically on pipeline success. For singleton modules (`shared-services`, `observability`, `network-core`), the apply job is `when: manual` to require deliberate human confirmation.

Destroy flow: Webhook-triggered modules (`vm-platform`, `k8s-platform`) accept a `TF_ACTION=destroy` variable. When set, the apply stage runs `tofu destroy -auto-approve` instead of `tofu apply`.

Listing 4.1 shows the canonical `.gitlab-ci.yml` for the `vm-platform` module. Singleton modules differ only in the apply job: `when: manual` replaces `when: on_success`, and there is no destroy branch.

```
image: ghcr.io/opentofu/opentofu:1.9

variables:
  TF_ACTION: "apply"
```

```
TF_STATE_NAME: "vm-vm-label"
TF_VAR_linode_token: "${LINODE_TOKEN}"

.tofu_init: &tofu_init
  before_script:
    - tofu init
      -reconfigure
      -backend-config="address=${CI_API_V4_URL}/projects/${
        ↳ INFRA_STATE_PROJECT_ID}/terraform/state/${TF_STATE_NAME}"
      -backend-config="lock_address=${CI_API_V4_URL}/projects/${
        ↳ INFRA_STATE_PROJECT_ID}/terraform/state/${TF_STATE_NAME}/lock"
      -backend-config="unlock_address=${CI_API_V4_URL}/projects/${
        ↳ INFRA_STATE_PROJECT_ID}/terraform/state/${TF_STATE_NAME}/lock"
      -backend-config="username=gitlab-ci-token"
      -backend-config="password=${CI_JOB_TOKEN}"
      -backend-config="lock_method=POST"
      -backend-config="unlock_method=DELETE"
      -backend-config="retry_wait_min=5"

  stages:
    - validate
    - plan
    - apply

  validate:
    stage: validate
    <<: *tofu_init
    script:
      - tofu validate
    rules:
      - if: $CI_PIPELINE_SOURCE == "trigger"

  plan:
    stage: plan
    <<: *tofu_init
    script:
      - tofu plan -out=tfplan
    artifacts:
      paths:
        - tfplan
      expire_in: 1 hour
    rules:
      - if: $CI_PIPELINE_SOURCE == "trigger"

  apply:
    stage: apply
    <<: *tofu_init
    script:
      - |
        if [ "${TF_ACTION}" = "destroy" ]; then
          tofu destroy -auto-approve
        else
          tofu apply -auto-approve tfplan
        fi
    dependencies:
      - plan
```

```
rules:
  - if: $CI_PIPELINE_SOURCE == "trigger" && $TF_ACTION == "destroy"
    when: always
  - if: $CI_PIPELINE_SOURCE == "trigger"
    when: on_success
```

Listing 4.1: .gitlab-ci.yml for vm-platform (webhook-triggered module)

Listing 4.2 shows a complete pipeline for singleton modules (`network-core`, `shared-services`, `observability`) where a human must confirm the apply step:

```
image: ghcr.io/opentofu/opentofu:1.9

variables:
  TF_STATE_NAME: "shared-prod"
  TF_VAR_linode_token: "${LINODE_TOKEN}"

.tofu_init: &tofu_init
  before_script:
    - tofu init
      -backend-config="address=${CI_API_V4_URL}/projects/${
        ↳ INFRA_STATE_PROJECT_ID}/terraform/state/${TF_STATE_NAME}"
      -backend-config="lock_address=${CI_API_V4_URL}/projects/${
        ↳ INFRA_STATE_PROJECT_ID}/terraform/state/${TF_STATE_NAME}/lock"
      -backend-config="unlock_address=${CI_API_V4_URL}/projects/${
        ↳ INFRA_STATE_PROJECT_ID}/terraform/state/${TF_STATE_NAME}/lock"
      -backend-config="username=gitlab-ci-token"
      -backend-config="password=${CI_JOB_TOKEN}"
      -backend-config="lock_method=POST"
      -backend-config="unlock_method=DELETE"
      -backend-config="retry_wait_min=5"

  stages:
    - validate
    - plan
    - apply

  validate:
    stage: validate
    <<: *tofu_init
    script:
      - tofu validate
    rules:
      - if: $CI_COMMIT_BRANCH == "main"

  plan:
    stage: plan
    <<: *tofu_init
    script:
      - tofu plan -out=tfplan
    artifacts:
      paths:
        - tfplan
      expire_in: 1 hour
    rules:
      - if: $CI_COMMIT_BRANCH == "main"
```

```
apply:
  stage: apply
  <<: *tofu_init
  script:
    - tofu apply -auto-approve tfplan
  dependencies:
    - plan
  rules:
    - if: $CI_COMMIT_BRANCH == "main"
      when: manual
```

Listing 4.2: .gitlab-ci.yml for singleton modules (manual apply)

Note

The single required CI/CD variable to set in the *infra-state* GitLab project group is `INFRA_STATE_PROJECT_ID` — the numeric project ID of the *infra-state* repository (visible in GitLab under **Settings** → **General**). `CI_JOB_TOKEN` and `CI_API_V4_URL` are provided automatically by GitLab CI. `LINODE_TOKEN` must be set with Read-/Write API access.

4.5 Webhook Integration with the Webshop

1. An Admin approves an order in the webshop (or places one directly).
2. For each webhook configured on the product-environment combination, the webshop sends:

```
POST <gitlab-webhook-url>
X-Gitlab-Token: <secret>
Content-Type: application/json

{
  "ref": "main",
  "variables": [
    {"key": "TF_VAR_vm_label", "value": "myvm-01"},
    {"key": "TF_VAR_vpc_id", "value": "12345"},
    ...
  ]
}
```

3. GitLab starts a pipeline for the `main` branch, injecting the variables as environment variables.
4. The pipeline reads `TF_VAR_*` variables, runs `tofu plan` then `tofu apply`.
5. On completion (success or failure), the webshop's polling goroutine detects the status change and updates the order record.

4.6 Dependency Between Modules

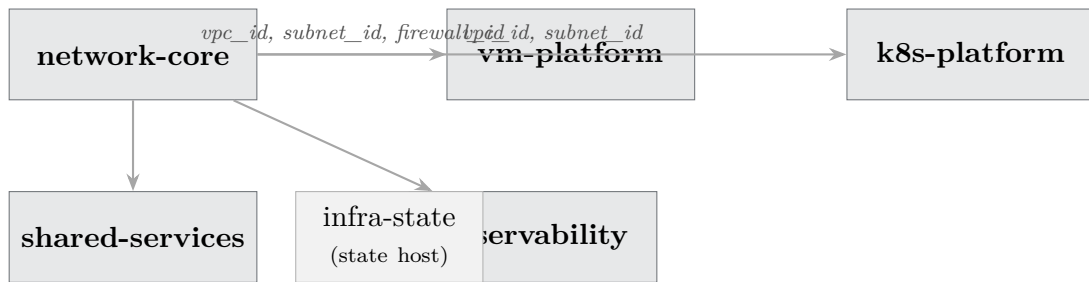


Figure 4.1: Module dependency graph

network-core must be applied first. Its outputs (*vpc_id*, *subnet_ids*, *firewall_id*) are passed as input variables to all other modules.

Chapter 5

OpenTofu Modules

5.1 network-core

Purpose: Foundational network infrastructure. Apply once per environment; no webhook trigger.

Resources created: `linode_vpc` (VPC), `linode_vpc_subnet` (subnets via `for_each`), `linode_firewall` (base inbound/outbound policy), `linode_domain` (optional DNS zone).

Key design choice: Subnets are defined as a `map(object({cidr, description}))` variable. This allows adding subnets by editing a single variable without changing the resource code.

Key outputs: `vpc_id`, `subnet_ids` (map: label $\rightarrow$ ID), `firewall_id`, `dns_zone_id`.

5.2 vm-platform

Purpose: Provision individual Linode VMs on demand. One state per VM (`vm-{label}`). Webhook-triggered by the webshop.

Resources created: `linode_firewall` (VM-specific), `linode_instance`, `linode_firewall_device` (attaches firewall to VM), `linode_volume` (optional block storage), `linode_domain_record` (optional A record).

Key design choice: The VM gets its own firewall (in addition to the base firewall from `network-core`) to allow per-VM port rules without affecting the shared base policy.

Destroy flow: Triggered by the webshop decommission action. `TF_ACTION=destroy` $\rightarrow$ `tofu destroy -auto-approve`.

5.3 k8s-platform

Purpose: Provision Linode Kubernetes Engine (LKE) clusters. One state per cluster (`k8s-{label}`). Webhook-triggered.

Resources created: `linode_firewall` (cluster-specific), `linode_lke_cluster` with dynamic `pool` blocks (supports multiple node pools with optional autoscaler), `linode_domain_record` (optional).

Key output: `kubeconfig` (base64-encoded, marked `sensitive`) — retrieve with `tofu output -raw kubeconfig | base64 -d`.

Node pool design: Node pools are a `list(object)` variable with an optional `autoscaler` sub-object, allowing any combination of fixed-size and auto-scaling pools in a single cluster.

5.4 shared-services

Purpose: Long-lived shared services: GitLab CE, HashiCorp Vault, container registry. Single state (`shared-prod`). Manual apply.

Resources created: Two `linode_instance` resources (GitLab, Vault), two `linode_volume` resources (data volumes), one `linode_firewall` with tailored inbound rules (HTTP/HTTPS/Git SSH for GitLab; Vault API 8200 restricted to RFC1918), two `linode_firewall_device` attachments, optional DNS A records.

Operational note: GitLab CE and Vault are installed via post-provisioning scripts (cloud-init or Ansible), not managed by OpenTofu. OpenTofu only provisions the infrastructure layer.

5.5 observability

Purpose: Grafana, Prometheus, and Loki on a single VM. Single state (`observability-prod`). Manual apply.

Resources created: `linode_instance`, `linode_volume` (200 GB default for metrics and logs), `linode_firewall` (Grafana port configurable, Prometheus/Loki restricted to RFC1918), optional DNS A record.

Key output: `grafana_url` — `http://{ip}:{port}`.

Part III

Administrator Guide

Chapter 6

Initial Setup & Installation

6.1 Prerequisites

- Docker 24+ (or a Kubernetes cluster)
- PostgreSQL 15+ (external or containerized)
- SMTP server credentials (optional, can be configured later)
- Microsoft Entra ID application registration for OIDC (optional for local-only operation)
- Linode API token with Read/Write scope
- GitLab instance with webhook support

6.2 Environment Variables

The webshop is configured entirely via environment variables. Copy `.env.example` to `.env` and populate the following:

Table 6.1: Required environment variables

Variable	Description	Required
<code>DATABASE_URL</code>	PostgreSQL DSN	Yes
<code>SESSION_KEY</code>	32-byte hex session encryption key	Yes
<code>ADMIN_EMAIL</code>	Initial Root email	Yes
<code>ADMIN_PASSWORD</code>	Initial Root password	Yes
<code>BASE_URL</code>	Public URL of the webshop	Yes
<code>OIDC_CLIENT_ID</code>	Entra ID application client ID	No
<code>OIDC_CLIENT_SECRET</code>	Entra ID client secret	No
<code>OIDC_ISSUER</code>	Entra ID issuer URL	No
<code>SMTP_HOST</code>	SMTP host (overridable in Admin UI)	No
<code>SMTP_PORT</code>	SMTP port (default 587)	No
<code>SMTP_FROM</code>	Sender address	No
<code>SMTP_USERNAME</code>	SMTP username	No
<code>SMTP_PASSWORD</code>	SMTP password	No
<code>AI_PROVIDER</code>	AI translation provider (startup only)	No

Note

SMTP and AI translation can also be configured (and overridden) through the Admin UI at `/admin/smtp` and `/admin/ai-config`. Database settings take precedence over environment variables at runtime.

6.3 Deployment on a Docker Host

This section describes a production-grade deployment on a single Linux host using Docker and an Nginx reverse proxy with TLS termination.

Architecture

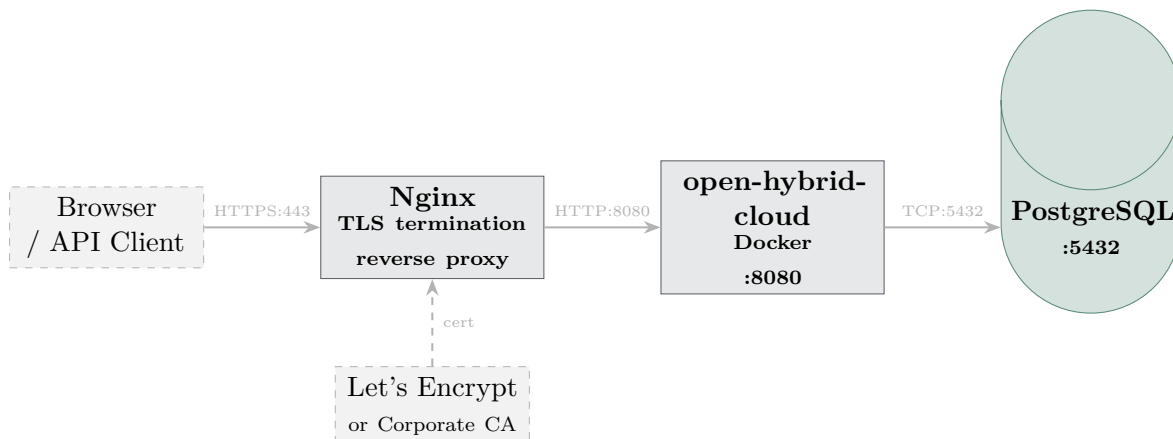


Figure 6.1: Docker Host deployment architecture

Step-by-Step Deployment Procedure

1. Install Docker and Docker Compose.

```
# Debian/Ubuntu
apt-get install -y docker.io docker-compose-v2
systemctl enable --now docker
```

2. Clone the repository and build the image.

```
git clone https://github.com/Maximilian-Maag/open-hybrid-cloud.git
cd open-hybrid-cloud
make docker-build
# or pull from Docker Hub:
docker pull maximilianmaag/open-hybrid-cloud:latest
```

3. Create the environment file.

```
cp infra/docker-host/.env.example infra/docker-host/.env
# Edit .env with a text editor; set at minimum:
# DATABASE_URL, SESSION_KEY, ADMIN_EMAIL, ADMIN_PASSWORD, BASE_URL
```

4. Start PostgreSQL and the application.

```
cd infra/docker-host
docker compose up -d
```

5. Run database migrations.

```
docker exec open-hybrid-cloud /migrate
```

6. Obtain a TLS certificate.

Option A — Let's Encrypt (public domain):

```
apt-get install -y certbot python3-certbot-nginx
certbot --nginx -d shop.example.com
```

Option B — Corporate CA (internal PKI): Place the signed certificate and key at the paths referenced in [infra/docker-host/nginx.conf](#) (see below) and restart Nginx.

7. Configure Nginx as TLS-terminating reverse proxy.

Listing 6.1 shows a minimal but production-ready Nginx configuration:

```
server {
    listen 80;
    server_name shop.example.com;
    return 301 https://$host$request_uri;
}

server {
    listen 443 ssl http2;
    server_name shop.example.com;
```

```
ssl_certificate    /etc/letsencrypt/live/shop.example.com/fullchain.pem;
ssl_certificate_key /etc/letsencrypt/live/shop.example.com/privkey.pem;
ssl_protocols      TLSv1.2 TLSv1.3;
ssl_ciphers        HIGH:!aNULL:!MD5;

client_max_body_size 10m;

location / {
    proxy_pass      http://127.0.0.1:8080;
    proxy_set_header Host          $host;
    proxy_set_header X-Real-IP     $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
    proxy_read_timeout 60s;
}
```

Listing 6.1: infra/docker-host/nginx.conf

8. Reload Nginx.

```
nginx -t && systemctl reload nginx
```

9. Verify the deployment.

Navigate to <https://shop.example.com> and log in with **ADMIN_EMAIL** / **ADMIN_PASSWORD**. Change the password immediately under **Settings** → **My Profile**.

10. (Optional) Enable automatic certificate renewal.

```
# Certbot installs a systemd timer automatically
systemctl status certbot.timer
```

Backup & Restore

```
# Backup PostgreSQL
docker exec webshop-postgres pg_dump -U webshop webshop \
  | gzip > backup-$(date +%Y%m%d).sql.gz

# Restore
zcat backup-20260101.sql.gz \
  | docker exec -i webshop-postgres psql -U webshop webshop
```

6.4 Deployment on Kubernetes

This section covers deploying the webshop to an existing Kubernetes cluster (e.g. the LKE cluster provisioned by k8s-platform).

Architecture

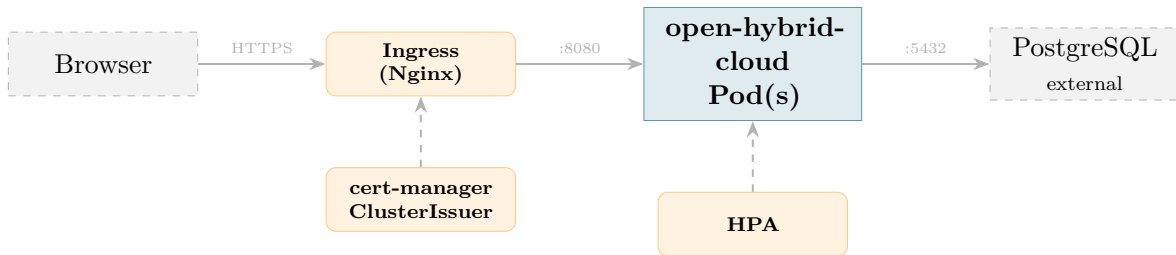


Figure 6.2: Kubernetes deployment architecture

Prerequisites

- `kubectl` and `helm` (v3.12+) configured for the target cluster
- Nginx Ingress Controller installed (`ingress-nginx`)
- `cert-manager` installed with a `ClusterIssuer` for Let's Encrypt or a corporate CA
- The image `maximilianmaag/open-hybrid-cloud:tag` is publicly available on Docker Hub — no `imagePullSecret` required

Deployment with Helm

The repository ships a Helm chart at [infra/helm/open-hybrid-cloud/](#). It covers the full deployment in a single command and handles database migrations automatically as a pre-install/pre-upgrade hook.

```

helm upgrade --install open-hybrid-cloud \
  infra/helm/open-hybrid-cloud \
  --namespace open-hybrid-cloud \
  --create-namespace \
  --set env.databaseUrl="postgres://user:pass@pg.example.com:5432/webshop?sslmode
↪ =require" \
  --set env.sessionSecret("<32-byte-hex>" \
  --set env.adminPassword("<initial-password>" \
  --set ingress.hosts[0].host=shop.example.com \
  --set ingress.tls[0].hosts[0]=shop.example.com \
  --set ingress.tls[0].secretName=open-hybrid-cloud-tls

```

Listing 6.2: Helm install with required values

Note

Sensitive values should be passed via a separate `secrets.values.yaml` file (not committed to version control) rather than as `--set` arguments, so they do not appear in shell history.

```

env:
  databaseUrl: "postgres://user:pass@pg.example.com:5432/webshop?sslmode=require"

```

```

sessionSecret: "
  ↪ aabbccddeeff00112233445566778899aabbccddeeff00112233445566778899"
adminPassword: "change-me-immediately"
entra:
  tenantId: "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxx"
  clientId: "xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxx"
  clientSecret: "your-client-secret"
  redirectUrl: "https://shop.example.com/auth/callback"

```

Listing 6.3: Example secrets.values.yaml (do not commit)

```

helm upgrade --install open-hybrid-cloud infra/helm/open-hybrid-cloud \
  --namespace open-hybrid-cloud --create-namespace \
  -f values.production.yaml \
  -f secrets.values.yaml

```

Table 6.2 lists the most important configurable values.

Table 6.2: Key Helm values

Key	Description	Default
image.repository	Docker Hub image name	maximilianmaag/open-hybrid-cloud
image.tag	Image tag (empty = appVersion)	""
replicaCount	Deployment replicas	2
env.databaseUrl	PostgreSQL DSN (required)	—
env.sessionSecret	Session encryption key (req.)	—
env.adminPassword	Initial admin password (req.)	—
ingress.hosts	Ingress host list	shop.example.com
ingress.tls	TLS configuration	(see values.yaml)
autoscaling.enabled	Enable HPA	false
autoscaling.maxReplicas	HPA max replica count	6
migrate.enabled	Run DB migrations as hook	true

Step-by-Step Manual Deployment (without Helm)

1. Create the namespace.

```
kubectl create namespace open-hybrid-cloud
```

2. (Optional) Create an image pull secret.

The container image `maximilianmaag/open-hybrid-cloud` is publicly available on Docker Hub and requires no authentication. Skip this step for standard deployments. Only required if you mirror the image to a private registry:

```
kubectl create secret docker-registry my-registry-secret \
  --namespace open-hybrid-cloud \
  --docker-server=<your-registry> \
  --docker-username=<username> \
  --docker-password=<password>
```

3. Create the application Secret.

```
apiVersion: v1
kind: Secret
metadata:
  name: open-hybrid-cloud-env
  namespace: open-hybrid-cloud
type: Opaque
stringData:
  DATABASE_URL: "postgres://webshop:password@pg.example.com:5432/webshop?
    ↪ sslmode=require"
  SESSION_KEY: "000102030405060708090
    ↪ a0b0c0d0e0f101112131415161718191a1b1c1d1e1f"
  ADMIN_EMAIL: "admin@example.com"
  ADMIN_PASSWORD: "changeme"
  BASE_URL: "https://shop.example.com"
```

Listing 6.4: Kubernetes Secret for the webshop

4. Run database migrations as a Kubernetes Job.

```
apiVersion: batch/v1
kind: Job
metadata:
  name: open-hybrid-cloud-migrate
  namespace: open-hybrid-cloud
spec:
  template:
    spec:
      restartPolicy: Never
      containers:
        - name: migrate
          image: maximilianmaag/open-hybrid-cloud:latest
          command: ["/migrate"]
          envFrom:
            - secretRef:
                name: open-hybrid-cloud-env
```

Listing 6.5: Migration Job

```
kubectl apply -f migrate-job.yaml
kubectl wait --for=condition=complete job/open-hybrid-cloud-migrate \
  -n open-hybrid-cloud --timeout=120s
```

5. Deploy the application.

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: open-hybrid-cloud
  namespace: open-hybrid-cloud
spec:
  replicas: 2
  selector:
    matchLabels:
      app: open-hybrid-cloud
  template:
    metadata:
      labels:
        app: open-hybrid-cloud
    spec:
      containers:
        - name: app
          image: maximilianmaag/open-hybrid-cloud:latest
          ports:
            - containerPort: 8080
          envFrom:
            - secretRef:
                name: open-hybrid-cloud-env
          livenessProbe:
            httpGet:
              path: /health
              port: 8080
            initialDelaySeconds: 10
            periodSeconds: 30
          readinessProbe:
            httpGet:
              path: /health
              port: 8080
            initialDelaySeconds: 5
            periodSeconds: 10
          resources:
            requests:
              cpu: "100m"
              memory: "128Mi"
            limits:
              cpu: "500m"
              memory: "512Mi"
---
apiVersion: v1
kind: Service
metadata:
```

```

name: open-hybrid-cloud
namespace: open-hybrid-cloud
spec:
  selector:
    app: open-hybrid-cloud
  ports:
    - port: 80
      targetPort: 8080

```

Listing 6.6: Deployment and Service

6. Configure the Ingress with TLS.

```

apiVersion: networking.k8s.io/v1
kind: Ingress
metadata:
  name: open-hybrid-cloud
  namespace: open-hybrid-cloud
  annotations:
    nginx.ingress.kubernetes.io/proxy-body-size: "10m"
    cert-manager.io/cluster-issuer: "letsencrypt-prod"
spec:
  ingressClassName: nginx
  tls:
    - hosts:
        - shop.example.com
      secretName: open-hybrid-cloud-tls
  rules:
    - host: shop.example.com
      http:
        paths:
          - path: /
            pathType: Prefix
            backend:
              service:
                name: open-hybrid-cloud
                port:
                  number: 80

```

Listing 6.7: Ingress with cert-manager TLS

7. (Optional) Configure Horizontal Pod Autoscaler.

```

apiVersion: autoscaling/v2
kind: HorizontalPodAutoscaler
metadata:
  name: open-hybrid-cloud
  namespace: open-hybrid-cloud
spec:
  scaleTargetRef:
    apiVersion: apps/v1
    kind: Deployment
    name: open-hybrid-cloud
  minReplicas: 2
  maxReplicas: 6
  metrics:

```

```
- type: Resource
  resource:
    name: cpu
    target:
      type: Utilization
      averageUtilization: 70
```

Listing 6.8: HorizontalPodAutoscaler

8. Apply all manifests and verify.

```
kubectl apply -f secret.yaml
kubectl apply -f deployment.yaml
kubectl apply -f ingress.yaml
kubectl apply -f hpa.yaml # optional

kubectl rollout status deployment/open-hybrid-cloud -n open-hybrid-cloud
kubectl get ingress -n open-hybrid-cloud
```

Note

The polling goroutines that track GitLab pipeline status run inside each pod. With multiple replicas, advisory locks in PostgreSQL (`pg_try_advisory_lock`) ensure that only one pod polls each pipeline at a time, preventing duplicate status updates.

6.5 Database Migrations

Migrations are versioned SQL files in `internal/migrations/`. Run migrations before starting the server:

```
# Run all pending migrations
make migrate

# Or directly
go run ./cmd/migrate
```

Migrations are idempotent (IF NOT EXISTS guards). Re-running is safe.

6.6 First Login

1. Navigate to the webshop URL.
2. Click **Local Login** (not **Sign in with Microsoft**).
3. Enter the email and password from `ADMIN_EMAIL` and `ADMIN_PASSWORD`.
4. Navigate to **Settings** → **My Profile** and change the default password immediately.

Chapter 7

System Configuration

7.1 GitLab Sources

Under **Administration** → **GitLab Sources**:

1. Click **Add New Source**.
2. Enter a **Name** (label), the instance **URL**, and a **Personal Access Token** with `read_api` scope.
3. Save. The source is now available when creating environments and when browsing repositories for parameter import.

7.2 Deployment Environments

Under **Administration** → **Deployment Environments**:

1. Click **Add New Environment**.
2. Enter a **Name** (e.g. *Linode Frankfurt Production*), select the **GitLab Source**, and enter the **Webhook URL** and **Webhook Token**.
3. Save. The environment becomes available when assigning products.

Note

The Webhook URL points to a specific GitLab project pipeline trigger endpoint, e.g.:

<https://gitlab.example.com/api/v4/projects/42/ref/main/trigger/pipeline>

7.3 SMTP Configuration

Under **Administration** → **Email** (</admin/smtp>):

Enter host, port, sender address, username, password, and TLS mode. Click **Save**. The notification service reloads credentials immediately without a server restart.

7.4 AI Translation

Under **Administration** → **AI Translation** (</admin/ai-config>):

1. Select a **Provider**: Claude, OpenAI, Azure OpenAI, Ollama, or LocalAI.
2. Enter the **API endpoint** and **API key**. For Ollama/LocalAI, the endpoint is a local URL and no key is required.
3. Enter the **model name** (e.g. `claude-opus-4-7`, `gpt-4o`, `llama3.1`).
4. Click **Save**. The translator instance is replaced immediately — the next translation request uses the new provider.

7.5 Branding

Under **Administration** → **Shop Design** (</admin/branding>):

- **Shop Name** and **Subtitle** — shown in the header.
- **Primary Color** — header, footer, navigation (default: `#131921`).
- **Accent Color** — buttons and call-to-action elements (default: `#febd69`).
- **Logo** — PNG or SVG, displayed in the header; replaces the shop name text if present.
- **Imprint** — HTML-formatted legal imprint text, shown at </impressum>.

All branding values are stored in a singleton database row and cached in-memory with a mutex for read performance.

7.6 Currencies

Under **Administration** → **Currencies**:

1. The **base currency** (default EUR) is the currency in which all product prices are stored.
2. Optionally configure an exchange rate API key and endpoint.
3. Click **Update Rates** to fetch live rates.
4. Individual rates can be overridden manually for fixed-rate environments.

Prices are displayed in the user's locale currency by looking up the browser locale and applying the stored rate.

Chapter 8

User Management

8.1 Roles

Table 8.1: User roles and capabilities

Role	Description	Auth method	DB role
Admin	Full access; direct ordering; approval	SSO	admin
Root	Catalog & config; no ordering	Local	root
Project Manager	Orders own infra; approval required	SSO	user

8.2 Creating a Local Account

Under **Administration** → **Users**:

1. Click **New User**.
2. Enter **Name**, **Email**, **Password**, and select a **Role**.
3. Click **Save**.

SSO users are created automatically on first login.

8.3 Editing Users

Click **Edit** next to a user to change name, email, or role. The password can only be changed by the user themselves under **Settings** → **My Profile**.

8.4 Deactivating Users

Click **Deactivate** on the user edit page. Deactivated users cannot log in. Their orders and projects remain visible in the system.

Chapter 9

Audit Log

The audit log under `/audit` is an append-only, tamper-evident record of all system actions.

9.1 Logged Events

Table 9.1: Audited actions

Action	Trigger
order_placed	User submits an order
order_approved	Admin approves
order_rejected	Admin rejects (with comment)
order_deployed	Pipeline completes successfully
order_failed	Pipeline fails
decommission_started	User triggers decommission
decommission_complete	Destroy pipeline succeeds
user_login	Successful authentication
user_created	Admin creates a user
config_changed	System configuration updated

9.2 Filtering

Use the filter form to narrow results by:

- **Date range** (from / to)
- **User**
- **Action type**

9.3 Export

Click **Export CSV** or **Export PDF** to download the currently filtered audit log for compliance reporting.

Part IV

Developer Guide

Chapter 10

Development Environment

10.1 Prerequisites

- Go 1.23+
- Node.js 20+ (for Tailwind CSS compilation)
- Docker & Docker Compose (for local PostgreSQL)
- `golangci-lint` (static analysis — installed via `make install-requirements`)
- `pdflatex` or `lualatex` (for handbook compilation, optional)

On Ubuntu or Manjaro all prerequisites except `pdflatex` can be installed with a single command:

```
make install-requirements
```

This detects the native package manager (`pacman` or `apt-get`), installs Go, Node.js, npm, Docker, and Docker Compose, and then installs the Go tools `goimports` and `golangci-lint` via `go install`. A re-login is required afterwards for the Docker group membership to take effect.

Note

Ubuntu note: The `golang-go` package in the Ubuntu repositories may lag behind the latest Go release. If Go 1.23+ is not available, install it manually from <https://go.dev/dl/>.

10.2 Quick Start

```
# Clone and enter the repo
git clone https://github.com/Maximilian-Maag/open-hybrid-cloud.git
cd open-hybrid-cloud

# Install all required tools (Ubuntu / Manjaro)
make install-requirements
```

```
# Copy and populate environment
cp .env.example .env
# Edit .env with your local DB credentials

# Start PostgreSQL and the server in one command
make dev

# In another terminal: CSS hot-reload
make css-watch
```

`make dev` starts Docker Compose (PostgreSQL), runs CSS build, then starts the Go server with live environment variables from `.env`.

10.3 Makefile Targets

Table 10.1: Makefile targets

Target	Action
<code>make build</code>	Build CSS + Go binary
<code>make run</code>	Start server (requires running DB)
<code>make migrate</code>	Run database migrations
<code>make css</code>	Build Tailwind CSS once
<code>make css-watch</code>	Watch and rebuild CSS on change
<code>make test</code>	Run all Go tests
<code>make vet</code>	Run <code>go vet</code>
<code>make lint</code>	Run <code>golangci-lint</code>
<code>make install-requirements</code>	Install system packages + Go tools (Ubuntu/Manjaro)
<code>make docker-build</code>	Build Docker image
<code>make dev</code>	Start DB + server (development)
<code>make dev-down</code>	Stop DB + server
<code>make clean</code>	Remove build artefacts
<code>make docs</code>	Compile the LaTeX handbook to PDF
<code>make e2e</code>	Run Playwright end-to-end tests (requires Docker)

The repository also includes a local smoke test runner in `cmd/smoke/`. Run:

```
DATABASE_URL=postgres://postgres:postgres@localhost:5432/infrashop?sslmode=
↪ disable \
go run ./cmd/smoke
```

This command creates temporary rows and verifies cascading cleanup for product deletion.

Chapter 11

Codebase Structure

11.1 Directory Layout

```
open-hybrid-cloud/
+-- cmd/
|   +-- server/          # main.go -- server entry point
|   +-- migrate/         # main.go -- migration runner
|   +-- smoke/           # main.go -- smoke test runner
+-- e2e/                 # Playwright end-to-end tests (src/e2e/)
+-- docs/                # Documentation (this handbook)
|   +-- handbook.tex
|   +-- handbook.pdf     # compiled output
+-- internal/
|   +-- aitranslation/   # AI translator interface + backends
|   +-- audit/           # AuditService implementation
|   +-- auth/            # OIDC, session store, password hashing
|   +-- config/          # Config struct loaded from env
|   +-- exchange/        # Currency exchange rate service
|   +-- handler/         # HTTP handlers and route registration
|   +-- i18n/            # 25-language key-value translation map
|   +-- migrations/      # SQL migration files (001_*.sql, ...)
|   +-- model/           # Shared domain model structs
|   +-- notification/    # SMTP notification service
|   +-- repository/      # Repository interfaces
|   |   +-- postgres/    # pgx/v5 implementations
|   +-- service/         # Service interfaces
|   +-- impl/            # Service implementations
+-- ui/
|   +-- comp/            # Shared templ component library
|   |   +-- layout.templ # Full-page layout shell + nav
|   |   +-- badge.templ  # Status badges
|   |   +-- button.templ # Button variants (Btn, BtnLink, IconBtn)
|   |   +-- card.templ   # Card, CardBody, CardHeader, PageHeader
|   |   +-- form.templ   # Field, Input, Textarea, Select, Checkbox
|   +-- pages/           # One package per route group
|   |   +-- admin/       # *.templ for all /admin/* pages
|   |   +-- approvals/
|   |   +-- audit/
|   |   +-- auth/
```

```
| +-- catalog/
| +-- home/
| +-- infra/
| +-- misc/
| +-- orders/
| +-- projects/
| +-- settings/
+-- static/
| +-- css/style.css      # Compiled Tailwind output
+-- ui.go                # embed.FS declarations
```

11.2 Layered Architecture

The code follows a strict layered architecture with downward-only dependencies:

Handler → Service → Repository → PostgreSQL

Handler layer ([internal/handler/](#)) HTTP handlers only. Read request, call service, render template. No business logic. No SQL.

Service layer ([internal/service/](#)) Business rules and orchestration. Calls repositories. Triggers notifications, audit writes, webhook dispatches. Defined as interfaces in [service.go](#), implemented in [impl/](#).

Repository layer ([internal/repository/](#)) Data access. Defined as interfaces (one per aggregate root). Implemented in [postgres/](#) using pgx. No business logic.

Model layer ([internal/model/](#)) Plain Go structs shared across layers. No methods beyond simple helpers.

11.3 Template System (templ)

All pages are authored as `.templ` files and compiled to Go by `templ generate ./...`. Each `.templ` file produces a corresponding `_templ.go` file which is committed alongside the source.

Page components receive a typed view struct:

```
// ui/pages/orders/orders.templ
templ OrderList(vd view.OrderListView) {
    @comp.Layout(vd.PageData) {
        // page content -- type-checked at compile time
    }
}
```

Handlers assemble the view struct and call `renderTempl`:

```
func (h *Handler) orderList(w http.ResponseWriter, r *http.Request) {
    orders, _ := h.orders.ListByUser(r.Context(), sess.UserID)
    renderTempl(w, r, orderpages.OrderList(view.OrderListView{
```

```

    PageData: h.buildPageData(w, r),
    Orders:  orders,
  )))
}

```

`buildPageData` assembles the common `view.PageData` struct providing:

- .Lang** User's active language code (e.g. "de").
- .Brand** Branding values (colours, shop name, logo flag).
- .Session** Current user session (name, role, email).
- .Flash** One-time flash message (success/error).
- .SearchQuery** Current catalog search term from the URL.

Translations are called directly in templ files:

```

{ i18n.T("orders.title", vd.Lang) } // translated string
{ string(o.Status) }                // named type requires cast

```

11.4 Design System

All UI is built from a small, self-contained component library under `src/ui/comp/`. Components are plain templ functions — no framework, no runtime dependencies.

Design Tokens

Two CSS custom properties drive every coloured surface:

var(--bp) Brand primary colour (default #131921, the dark header). Used for header, navigation, footer, active highlights, and price text.

var(--bs) Brand secondary / accent colour (default #febd69, amber). Used for CTA buttons, active language badges, and focus rings.

The values are injected by `comp.Layout` as inline CSS custom properties on `<html>`. Every component should reference these variables rather than hardcoding colours:

```

<!-- correct -->
<div style="background-color:var(--bp)">...</div>
<a style="color:var(--bs)">...</a>

<!-- wrong: breaks custom branding -->
<div class="bg-[#131921]">...</div>

```

Layout Shell (`comp.Layout`)

Every page is wrapped in `comp.Layout`:

```
templ MyPage(vd view.MyView) {  
    @comp.Layout(vd.PageData) {  
        // page content here  
    }  
}
```

`comp.Layout` renders: `<head>` with meta tags and stylesheets, sticky header with brand logo / search bar / user menu / language picker, the top navigation bar with role-based links, a flash banner slot, the `<main>` content area (2XL max-width, horizontal padding), and the footer. Nothing else should set `<html>`, `<head>`, or `<body>`.

Page Structure

`comp.PageHeader(title, backHref)` Full-width page title row with an optional back arrow link. Use at the top of every dedicated page.

`comp.PageHeaderAction(title, backHref, action)` Same as above but with an action slot on the right (e.g. a primary button).

`comp.Section(title)` A titled content group with a subtle separator. Use to divide a page into logical parts.

`comp.EmptyState(message)` Centred placeholder message for empty lists or zero results.

```
@comp.PageHeaderAction("Bestellungen", "/", action) {  
    @comp.BtnLink("Neue Bestellung", "/catalog",  
        comp.VPrimary, comp.SzMD)  
}  
@comp.Section("Offene Bestellungen") {  
    if len(orders) == 0 {  
        @comp.EmptyState("Keine Bestellungen vorhanden.")  
    }  
    // list rows...  
}
```

Cards

`comp.Card()` White surface with border and subtle shadow. Use as a generic content container.

`comp.CardBody()` Adds consistent padding inside a `Card`.

`comp.CardHeader(title)` A titled header strip for a card.

```
@comp.Card() {  
    @comp.CardHeader("Details") {}  
    @comp.CardBody() {  
        // form fields, table, prose, etc.  
    }  
}
```

Buttons

comp.Btn(label, variant, size, templ.Attributes) A `<button>` element. Pass extra HTML attributes (e.g. `hx-post`, `disabled`) via the fourth argument.

comp.BtnLink(label, href, variant, size) An `<a>` element styled as a button. Use for navigation CTAs.

comp.IconBtn(variant, size, templ.Attributes) Square icon button; children supply the icon SVG.

comp.BtnClass(variant, size) Returns just the Tailwind class string. Use when you need to apply button styling to an element not covered by the components above (e.g. an HTMX-powered `<button>` inside a `templ.children...` slot).

Variants:

<code>comp.VPrimary</code>	Filled, uses <code>var(-bp)</code> background — main CTAs.
<code>comp.VSecondary</code>	Outlined border button — secondary actions.
<code>comp.VGhost</code>	Transparent, text-only — low-emphasis actions.
<code>comp.VDanger</code>	Red filled — destructive actions (delete, reject).

Sizes:

<code>comp.SzXS</code>	Extra small — inline or table-row actions.
<code>comp.SzSM</code>	Small — secondary actions, compact layouts.
<code>comp.SzMD</code>	Medium (default) — primary page-level CTAs.

```
// Primary CTA
@comp.BtnLink("Jetzt bestellen", "/catalog/"+id,
              comp.VPrimary, comp.SzMD)

// Danger action with HTMX
@comp.Btn("Ablehnen", comp.VDanger, comp.SzSM,
          templ.Attributes{"hx-delete": "/orders/"+id,
                           "hx-confirm": "Wirklich ablehnen?"})

// Icon button (custom SVG inside)
@comp.IconBtn(comp.VGhost, comp.SzXS,
              templ.Attributes{"aria-label": "Bearbeiten"}) {
  <svg .../>
}
```

Badges

comp.Badge(label, color) A small inline pill. Use `comp.BadgeGreen`, `comp.BadgeRed`, `comp.BadgeYellow`, `comp.BadgeBlue`, or `comp.BadgeGray` for the color argument.

comp.StatusBadge(status) Convenience wrapper that maps an order/resource status string to the appropriate colour via `comp.StatusBadgeColor`. No need to branch on status in the template.

```
// Generic badge
@comp.Badge("Aktiv", comp.BadgeGreen)
```

```
@comp.Badge("Fehler", comp.BadgeRed)

// Status-aware badge (recommended for orders)
@comp.StatusBadge(string(order.Status))
```

Status colour mapping (via `comp.StatusBadgeColor`):

active, approved, completed	Green
pending	Yellow
failed, rejected	Red
provisioning, decommissioning	Blue
Everything else	Gray

Forms

All form fields use the components in `ui/comp/form.templ` for consistent label alignment, spacing, and required-marker rendering.

`comp.Field(label, name, required)` A `<label>` + slot wrapper. Place an `Input`, `Select`, or `Textarea` inside it.

`comp.Input(name, type, placeholder, value, attrs)` Styled `<input>`. Pass extra attributes (e.g. `hx-get`, `min`, `step`) via the last argument.

`comp.Textarea(name, placeholder, value, rows, attrs)` Styled `<textarea>`.

`comp.Select(name, attrs)` Styled `<select>` wrapper; children are `<option>` elements written inline.

`comp.Checkbox(name, label, checked)` A labelled checkbox with correct `for/id` wiring.

```
<form method="POST" action="/admin/products">
  @comp.Field("Produktname", "name", true) {
    @comp.Input("name", "text", "z.B. VM-Plattform",
               product.Name, nil)
  }
  @comp.Field("Beschreibung", "description", false) {
    @comp.Textarea("description", "", product.Description,
                  4, nil)
  }
  @comp.Field("Kategorie", "category_id", true) {
    @comp.Select("category_id", nil) {
      for _, c := range categories {
        <option value={ fmt.Sprint(c.ID) }
          selected?={ c.ID == product.CategoryID }>
          { c.Name }
        </option>
      }
    }
  }
  @comp.Field("Aktiv", "active", false) {
    @comp.Checkbox("active", "Produkt ist sichtbar",
                  product.Active)
```

```

    }
    @comp.Btn("Speichern", comp.VPrimary, comp.SzMD, nil)
</form>

```

Flash Messages

Flash messages are set server-side via `redirectWithFlash` and rendered automatically by `comp.Layout` from `pd.Flash`. HTMX responses that want to show a flash message use the `HX-Trigger` response header instead:

```

// In a handler that returns an HTMX partial:
w.Header().Set("HX-Trigger", `{"showFlash":{"kind":"success",
    "message":"Gespeichert"}}`)

```

Component Conventions

- **Never hardcode colours.** Always use `var(-bp)`, `var(-bs)`, or Tailwind's neutral palette (`slate-*`). Hardcoded hex values break custom branding.
- **Use `comp.VPrimary` for the single most important action** on a page. Use `comp.VSecondary` or `comp.VGhost` for everything else.
- **No JavaScript.** Interactive disclosure (menus, modals, accordions) uses native `<details>/<summary>`. Server-side partial updates use HTMX attributes. Never add `<script>` tags to component files.
- **No inline style except for brand tokens.** Arbitrary colours, sizes, and spacing belong in Tailwind utility classes, not `style=""` attributes.
- **Prefer `comp.StatusBadge` over manual branching.** The mapping is maintained centrally; adding a new status only requires updating `comp.StatusBadgeColor`.

11.5 Adding Translations

All UI strings live in `internal/i18n/i18n.go` as a `map[string]map[string]string`. To add a new key:

```

// In translations map:
"my.new.key": {
    "en": "English text",
    "de": "Deutscher Text",
    "fr": "Texte francais",
    // ... all 25 language codes
},

```

Use in templates with `{{t "my.new.key" .Lang}}`.

11.6 Database Migrations

Migration files follow the naming pattern `NNN_description.sql` and are executed in alphabetical order. Write migrations defensively:

```
ALTER TABLE products
  ADD COLUMN IF NOT EXISTS image_mime TEXT NOT NULL DEFAULT 'image/jpeg';

CREATE TABLE IF NOT EXISTS app_config (
  id INT PRIMARY KEY DEFAULT 1 CHECK (id = 1),
  smtp_host TEXT NOT NULL DEFAULT ''
  -- ...
);
```

Important

Never modify a migration that has already been applied to production. Always create a new migration file for schema changes.

Chapter 12

Testing

The project uses three complementary test layers. All three are invoked automatically in CI; locally the most common commands are `make test` (fast, no Docker needed) and `make e2e` (full browser stack, requires Docker).

12.1 Unit Tests

Unit tests live next to the code they test (`*_test.go`) and use Go's standard `testing` package. They cover:

- **Service layer** (`internal/service/impl/`) — all service methods are tested with in-memory stubs that implement the repository interfaces. No database, no network.
- **Handler helpers** (`internal/handler/`) — language detection (`parseAcceptLanguage`, `detectLang`) and the login rate-limiter are tested with direct function calls.
- **Polling worker** (`internal/polling/`) — `extractProjectID` is tested with table-driven cases.

Run unit tests:

```
make test      # runs go test ./...
make vet       # go vet (zero warnings required)
make lint     # golangci-lint
```

12.2 Integration Tests (testcontainers-go)

Repository-layer tests in `internal/repository/postgres/` spin up a real **PostgreSQL 16 Alpine** container via `testcontainers-go` and run all migrations before the first test.

TestMain

```
// testmain_test.go
func TestMain(m *testing.M) {
    ctr, err := tcpostgres.Run(ctx, "postgres:16-alpine",
        tcpostgres.WithDatabase("testdb"),
        testcontainers.WithWaitStrategy(
            wait.ForLog("database system is ready to accept connections").
                WithOccurrence(2).WithStartupTimeout(60*time.Second),
        ),
    )
    // runs migrations, opens pgxpool
    os.Exit(m.Run())
}
```

Test isolation

Every test that writes data calls `resetDB(t)` first, which truncates all data tables and restarts identity sequences:

```
func resetDB(t *testing.T) {
    _, err := testPool.Exec(ctx,
        "TRUNCATE audit_log, orders, products, ... RESTART IDENTITY CASCADE")
}
```

Fixture helper

Tests that need FK-dependent data call `newFixture(t)`, which creates the full dependency chain in one call:

```
gitlab_source -> deployment_environment -> user
               -> category -> product -> project
```

Integration tests require Docker; if the container cannot start the test binary exits with code 0 (skip, not fail) so CI stays green on environments without Docker.

12.3 End-to-End Tests with Playwright

End-to-end tests live in [src/e2e/](#) and drive a real **Chromium** browser via [playwright-go](#). The full application stack — PostgreSQL, all migrations, all service wiring — is started in-process for each test run.

Architecture

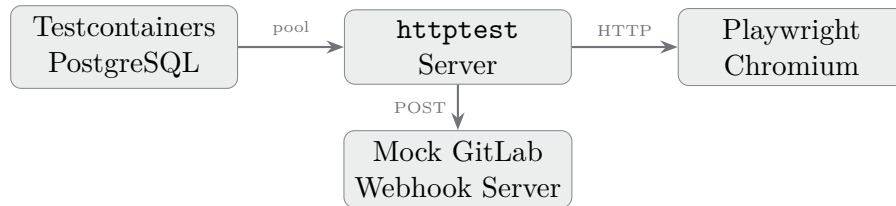


Figure 12.1: E2E test infrastructure

Testcontainers PostgreSQL A real `postgres:16-alpine` container is started by `TestMain`. All 12 migrations run against it and the admin user is bootstrapped (email `admin@e2etest.local`, password `e2eAdmin123!`).

`app.New()` The server wiring function from [internal/app/wire.go](#) assembles the full dependency graph (repos, services, handlers) and returns an `http.Handler`.

`httptest.NewServer` Wraps the handler in a local HTTP server on a random port. Playwright's `BaseURL` is set to this address.

Mock GitLab webhook server Approving an order triggers a GitLab pipeline via HTTP POST. A second `httptest.Server` intercepts these calls and returns `{"id":42}` so the approval flow completes without a real GitLab instance.

Chromium (headless) One browser is launched for the entire test run. Each test gets its own *browser context* (isolated cookie jar) via `newPage(t)`.

Test file overview

Table 12.1: E2E test files

File	Tests	What is verified
auth_test.go	3	Login success, invalid credentials, logout dropdown
approval_test.go	3	Empty list, Approve button, Reject button
catalog_test.go	3	Page loads, empty state, product navigation

Shared helpers

`newPage(t)` Creates a fresh browser context and page; the context is closed in `t.Cleanup`. Each test gets an empty cookie jar, so sessions never leak between tests.

`loginAs(t, page, email, password)` Navigates to `/login`, fills both fields, clicks `Submit`, and waits for the redirect to `/`.

`resetData(t)` Truncates all data tables (except `users`) with `RESTART IDENTITY CASCADE`, so the bootstrapped admin user survives across tests.

seedPendingOrder(t) Inserts the full FK chain (gitlab_source, environment, category, product, project, order) with **status = 'pending_approval'**. Uses **webhookBaseUrl** so the mock server intercepts the approval webhook call.

Key selector conventions

Because approvals use plain HTML `<form>` elements (no HTMX), tests use stable **action-attribute selectors**:

```
// approve button for a specific order
approveForm := page.Locator(
    fmt.Sprintf('form[action="/approvals/%d/approve"]', fix.orderID))
approveForm.Locator('button[type="submit"]').Click()

// reject form is inside a <details> -- open summary first
page.Locator('details.mt-4 > summary').First().Click()
noteInput := page.Locator('input[name="note"]')
noteInput.WaitFor()           // waits until visible
noteInput.Fill("reason")
```

After a form submit the 303 redirect reloads the same URL. `WaitForLoadState(NetworkIdle)` is used instead of `WaitForURL` (which would return immediately because the URL does not change).

Running the tests

```
# Headless (CI / default)
make e2e

# Headed -- watch the browser while tests run
E2E_HEADED=true make e2e

# First run downloads Chromium (~170 MB) into ~/.cache/ms-playwright/
```

If Docker is unavailable the test binary exits with code 0 (skip, not fail), consistent with the integration test policy.

Adding a new E2E test

1. Call `resetData(t)` at the start if the test writes data.
2. Use `newPage(t)` to get an isolated page.
3. Use `loginAs` to authenticate if needed.
4. Prefer `form[action="..."]` and `input[name="..."]` selectors over CSS classes (classes change; actions rarely do).
5. After a form submit call `page.WaitForLoadState(NetworkIdle)` before asserting.

Chapter 13

Database Schema

The schema is built by 12 sequential migrations (see Section 11.6). The table below shows the **current effective schema** — i.e. the cumulative result of all migrations.

13.1 Table Overview

Table 13.1: All database tables

Table	Purpose
users	User accounts (local + SSO) with role-based access
categories	Product catalogue categories
products	Infrastructure products offered in the catalogue
product_translations	Multi-language name + description per product
parameters	Configurable inputs for global/category/product scope
gitlab_sources	GitLab instance credentials (URL + access token)
deployment_environments	Named environments linked to a GitLab source
product_environments	Junction: which products are available in which environment (with price)
product_webhooks	Optional per-product pipeline trigger overrides
cost_centers	Billing cost centres referenced by projects and orders
projects	User-owned projects that group orders
orders	Individual infrastructure provisioning requests
infrastructure_elements	Deployed instances tracked per order
audit_log	Append-only log of significant user actions
exchange_rates	Cached currency conversion rates
branding	Single-row branding config (logo, colours, texts)
app_config	Single-row application config (SMTP, AI)

13.2 Detailed Schema

Each table is described with its columns (post all migrations), primary key, foreign keys, and notable constraints.

users

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
email	TEXT	—	NOT NULL, UNIQUE
name	TEXT	—	NOT NULL
role	TEXT	—	CHECK: admin, project_leader, root
sso_sub	TEXT	NULL	UNIQUE; set for Entra ID accounts
password_hash	TEXT	NULL	set for local accounts
active	BOOLEAN	TRUE	soft-disable without deletion
created_at	TIMESTAMPTZ	NOW()	

Constraint: sso_sub IS NOT NULL OR password_hash IS NOT NULL (every account must have at least one credential).

categories

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
name	TEXT	—	NOT NULL
display_order	INT	0	controls sidebar sort order

products

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
category_id	BIGINT	—	NOT NULL, FK → categories(id) ON DELETE CASCADE
base_language	TEXT	'de'	fallback language for translations
image	BYTEA	NULL	product image stored inline
created_at	TIMESTAMPTZ	NOW()	

product_translations

Column	Type	Default	Notes
product_id	BIGINT	—	PK, FK → <code>products(id)</code> ON DELETE CASCADE
language_code	TEXT	—	PK, e.g. <code>de</code> , <code>en</code> , <code>fr</code>
name	TEXT	—	NOT NULL
description	TEXT	"	NOT NULL

parameters

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
scope	TEXT	—	CHECK: <code>global</code> , <code>category</code> , <code>product</code>
scope_id	BIGINT	0	ID of the scoped entity (0 for global)
environment_id	BIGINT	0	0 = applies to all environments
name	TEXT	—	NOT NULL; variable name passed to pipeline
type	TEXT	—	CHECK: <code>string</code> , <code>number</code> , <code>bool</code> , <code>dropdown</code>
description	TEXT	"	
default_value	TEXT	"	
required	BOOLEAN	FALSE	
sensitive	BOOLEAN	FALSE	renders as <code>type="password"</code> in UI

gitlab_sources

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
name	TEXT	—	NOT NULL; human-readable label
url	TEXT	—	NOT NULL; GitLab base URL
access_token	TEXT	—	NOT NULL; stored in plain text

deployment_environments

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
name	TEXT	—	NOT NULL
description	TEXT	"	
gitlab_source_id	BIGINT	—	NOT NULL, FK → <code>gitlab_sources(id)</code>
webhook_url	TEXT	—	NOT NULL; GitLab trigger URL
webhook_token	TEXT	—	NOT NULL; pipeline trigger token

product_environments

Junction table linking products to environments with pricing.

Column	Type	Default	Notes
product_id	BIGINT	—	PK, FK → products(id) CASCADE
environment_id	BIGINT	—	PK, FK → deployment_environments(id)
price	NUMERIC(12,2)	0	monthly price
currency	TEXT	'EUR'	ISO 4217 code
cost_center_mode	TEXT	'project'	CHECK: project, select, overhead
forced_cost_center	BOOLEAN	FALSE	locks cost centre selection in order form

product_webhooks

Optional per-product pipeline overrides; if rows exist for a product+environment pair they replace the environment's default `webhook_url`. Multiple rows are fired in ascending `exec_order`.

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
product_id	BIGINT	—	NOT NULL, FK → products(id) CASCADE
environment_id	BIGINT	—	NOT NULL, FK → deployment_environments(id)
name	TEXT	—	NOT NULL; descriptive label
webhook_url	TEXT	—	NOT NULL
webhook_token	TEXT	—	NOT NULL
exec_order	INT	0	ascending; ties fire concurrently

cost_centers

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
code	TEXT	—	NOT NULL, UNIQUE; billing code
name	TEXT	—	NOT NULL
active	BOOLEAN	TRUE	inactive centres are hidden in UI

projects

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
name	TEXT	—	NOT NULL
description	TEXT	"	
owner_id	BIGINT	—	NOT NULL, FK → users(id)
cost_center_id	BIGINT	NULL	FK → cost_centers(id)
created_at	TIMESTAMPTZ	NOW()	

orders

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
project_id	BIGINT	—	NOT NULL, FK → <code>projects(id)</code> CASCADE
product_id	BIGINT	—	NOT NULL, FK → <code>products(id)</code> CASCADE
environment_id	BIGINT	—	NOT NULL, FK → <code>deployment_environments(id)</code>
user_id	BIGINT	—	NOT NULL, FK → <code>users(id)</code>
status	TEXT	'pending_approval'	see lifecycle below
parameters	JSONB	'{'	key/value map; passed to pipeline
cost_center_id	BIGINT	NULL	FK → <code>cost_centers(id)</code>
rejection_note	TEXT	"	filled on rejection
pipeline_id	JSONB	'[]'	array of GitLab pipeline IDs
created_at	TIMESTAMPTZ	NOW()	
updated_at	TIMESTAMPTZ	NOW()	

Status lifecycle: `pending_approval` → `approved` / `rejected` → `provisioning` → `completed` / `failed` → `decommissioning` → `decommissioned`.

infrastructure_elements

One row per deployed pipeline; created when an order is approved.

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
order_id	BIGINT	—	NOT NULL, FK → <code>orders(id)</code>
project_id	BIGINT	—	NOT NULL, FK → <code>projects(id)</code> CASCADE
environment_id	BIGINT	—	NOT NULL, FK → <code>deployment_environments(id)</code>
product_id	BIGINT	—	NOT NULL, FK → <code>products(id)</code> CASCADE
status	TEXT	'provisioning'	mirrors order status values
parameters	JSONB	'{'	copy of order parameters at deploy time
pipeline_id	JSONB	'[]'	tracked pipeline IDs
outputs	JSONB	'{'	key/value map of OpenTofu outputs
deployed_at	TIMESTAMPTZ	NOW()	

audit_log

Append-only; rows are never updated or deleted.

Column	Type	Default	Notes
id	BIGSERIAL	—	PK
user_id	BIGINT	—	NOT NULL, FK → users(id)
action	TEXT	—	NOT NULL; e.g. order.created
entity_id	BIGINT	—	NOT NULL; PK of the referenced entity
details	TEXT	"	extra context (e.g. rejection note)
created_at	TIMESTAMPTZ	NOW()	

exchange_rates

Column	Type	Default	Notes
currency_code	TEXT	—	PK; ISO 4217, e.g. EUR
rate	NUMERIC(18,6)	—	NOT NULL; relative to base currency (EUR = 1.0)
updated_at	TIMESTAMPTZ	NOW()	refreshed via POST /admin/currencies/refresh

branding

Singleton row (id = 1). Written via [POST /admin/branding](#).

Column	Type	Default	Notes
id	INT	1	PK, CHECK (id = 1)
logo_data	BYTEA	NULL	uploaded logo image
logo_mime	TEXT	'image/png'	MIME type of the logo
primary_color	TEXT	'#131921'	var(-bp) CSS token
secondary_color	TEXT	'#febd69'	var(-bs) CSS token
shop_name	TEXT	"	overrides APP_NAME env var
shop_subtitle	TEXT	"	overrides APP_SUBTITLE env var
imprint_text	TEXT	"	rendered on /impressum

app_config

Singleton row (id = 1). Written via [POST /admin/smtp](#) and [POST /admin/ai-config](#). DB values override environment variables at runtime.

Column	Type	Default	Notes
id	INT	1	PK, CHECK (id = 1)
smtp_host	TEXT	"	SMTP server hostname
smtp_port	TEXT	'587'	
smtp_from	TEXT	"	sender address
smtp_username	TEXT	"	
smtp_password	TEXT	"	stored in plain text
smtp_tls	BOOLEAN	FALSE	enable STARTTLS
ai_provider	TEXT	"	e.g. <code>claude</code> , <code>openai</code>
ai_endpoint	TEXT	"	custom API base URL
ai_api_key	TEXT	"	stored in plain text
ai_model	TEXT	"	model ID string

13.3 Foreign Key Relationships

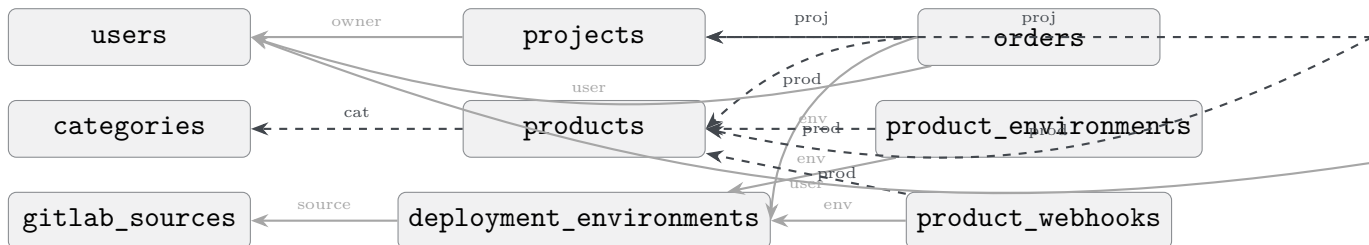


Figure 13.1: Simplified FK diagram. Dashed arrows = ON DELETE CASCADE; solid arrows = restricted.

Note

The cascade policy ensures that deleting a **product**, **project**, or **category** automatically removes all dependent rows (orders, infra elements, translations, etc.) without requiring explicit cleanup in application code.

13.4 Indexes

Table 13.2: Non-primary-key indexes

Index name	Table	Columns
idx_products_category	products	category_id
idx_orders_user	orders	user_id
idx_orders_status	orders	status
idx_orders_project	orders	project_id
idx_infra_project	infrastructure_elements	project_id
idx_infra_status	infrastructure_elements	status
idx_projects_owner	projects	owner_id
idx_audit_user	audit_log	user_id
idx_audit_action	audit_log	action
idx_parameters_scope	parameters	scope, scope_id, environment_id
idx_product_webhooks	product_webhooks	product_id, environment_id, exec_order

Chapter 14

Adding a New Feature

This chapter describes the canonical pattern for adding a feature end-to-end.

14.1 Example: Adding a New Admin Page

Goal: Add an admin page at `/admin/widgets` that lists and creates widgets stored in a new `widgets` database table.

14.1.1 Step 1: Create the Migration

Create `internal/migrations/010_widgets.sql`:

```
CREATE TABLE IF NOT EXISTS widgets (  
  id BIGSERIAL PRIMARY KEY,  
  name TEXT NOT NULL,  
  active BOOLEAN NOT NULL DEFAULT TRUE,  
  created_at TIMESTAMPTZ NOT NULL DEFAULT now()  
);
```

14.1.2 Step 2: Add the Model

In `internal/model/model.go`:

```
type Widget struct {  
  ID      int64  
  Name    string  
  Active  bool  
  CreatedAt time.Time  
}
```

14.1.3 Step 3: Define the Repository Interface

In `internal/repository/repository.go`:

```
type WidgetRepository interface {
    FindAll(ctx context.Context) ([]model.Widget, error)
    Create(ctx context.Context, name string) error
    Delete(ctx context.Context, id int64) error
}
```

14.1.4 Step 4: Implement the Repository

Create `internal/repository/postgres/widget.go`:

```
type widgetRepo struct{ pool *pgxpool.Pool }

func NewWidgetRepository(pool *pgxpool.Pool) repository.WidgetRepository {
    return &widgetRepo{pool: pool}
}

func (r *widgetRepo) FindAll(ctx context.Context) ([]model.Widget, error) {
    rows, err := r.pool.Query(ctx,
        'SELECT id, name, active, created_at FROM widgets ORDER BY name')
    // ... scan rows into []model.Widget
}
```

14.1.5 Step 5: Add the Handler

Add to the Handler struct in `routes.go`:

```
widgets repository.WidgetRepository
```

Add to Deps struct and wire in `New()`. Add a view struct in `internal/view/pages.go`:

```
type AdminWidgetsView struct {
    view.PageData
    Widgets []model.Widget
}
```

Add handler methods to a new file `handler/widgets.go`:

```
func (h *Handler) adminWidgets(w http.ResponseWriter, r *http.Request) {
    items, _ := h.widgets.FindAll(r.Context())
    renderTempl(w, r, adminpages.AdminWidgets(view.AdminWidgetsView{
        PageData: h.buildPageData(w, r),
        Widgets: items,
    }))
}
```

Register routes in `RegisterRoutes`:

```
mux.Handle("GET /admin/widgets", admin(h.adminWidgets))
mux.Handle("POST /admin/widgets", admin(h.adminWidgetCreate))
```

14.1.6 Step 6: Create the templ Component

Create `ui/pages/admin/widgets.templ`:

```
package admin

import (
    "github.com/Maximilian-Maag/open-hybrid-cloud/src/internal/i18n"
    "github.com/Maximilian-Maag/open-hybrid-cloud/src/internal/view"
    "github.com/Maximilian-Maag/open-hybrid-cloud/src/ui/comp"
)

templ AdminWidgets(vd view.AdminWidgetsView) {
    @comp.Layout(vd.PageData) {
        @comp.PageHeader(i18n.T("admin.widgets", vd.Lang), "/admin")
        @comp.Card() {
            <table class="w-full text-sm">
                <tbody>
                    for _, w := range vd.Widgets {
                        <tr class="hover:bg-slate-50">
                            <td class="px-4 py-3">{ w.Name }</td>
                        </tr>
                    }
                </tbody>
            </table>
        }
    }
}
```

Run `templ generate ./...` to compile to Go, then `go build ./...` to verify type correctness before running.

14.1.7 Step 7: Add i18n Keys

In `internal/i18n/i18n.go`, add the key with all 25 language translations.

14.1.8 Step 8: Wire Up in main.go

```
widgetRepo := pgRepo.NewWidgetRepository(pool)
// ...
h := handler.New(handler.Deps{
    // ...
    Widgets: widgetRepo,
})
```

14.1.9 Step 9: Test

Add test stubs in `internal/repository/` (or use the real DB in integration tests).
Run:

```
make test
make vet
make lint
```

Part V

Project Manager & Product Lifecycle Guide

Chapter 15

Ordering Infrastructure

15.1 The Order Lifecycle

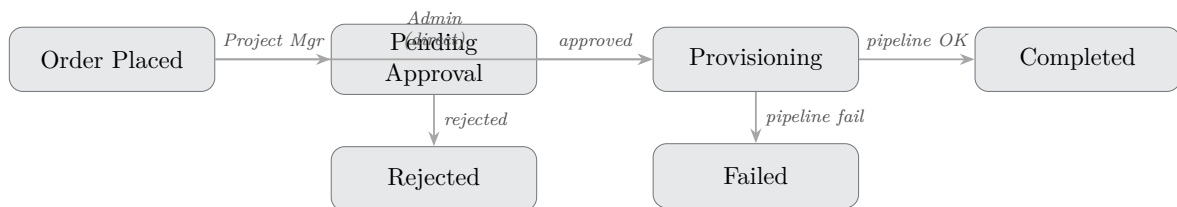


Figure 15.1: Order lifecycle state machine

15.2 Placing an Order

1. Navigate to **Catalog** and select a product.
2. Click **Configure** → select the deployment environment.
3. Select or create a **Project**.
4. Fill in the **Parameters** (required fields are marked).
5. Select or confirm the **Cost Center**.
6. Click **Deploy Now**.

As a **Project Manager** the order status changes to *Pending Approval*. An Admin receives an email notification.

As a **Admin** the GitLab webhook fires immediately.

15.3 Tracking Status

After placing an order, navigate to **My Orders** → **{order}**. The status updates live via HTMX polling every 30 seconds.

15.4 Using an Existing Deployment as a Template

1. Open **Infrastructure**.
2. Find the resource and click **Use as Template**.
3. The order form opens pre-filled with the original parameters.
4. Modify as needed and submit.

15.5 Decommissioning

1. Open **Infrastructure**.
2. Find the resource and click **Decommission**.
3. Confirm the action.
4. The destroy webhook fires; status changes to *Decommissioning*.
5. On pipeline completion, status becomes *Decommissioned*.

Cascade Decommissioning on Deletion

Deleting a **project**, **product**, or **category** automatically triggers decommissioning for all active infrastructure elements associated with that object before the database record is removed:

- **Project deleted** — all active infrastructure elements of the project receive a destroy webhook call.
- **Product deleted** — all active infrastructure elements provisioned from that product receive a destroy webhook call.
- **Category deleted** — all active infrastructure elements of every product in the category receive a destroy webhook call.

Infrastructure already in status *Decommissioning* or *Decommissioned* is skipped. Dependent database rows (translations, parameters, environment assignments) are removed automatically via PostgreSQL ON DELETE CASCADE constraints after the webhooks have been triggered.

Important

Deleting a project, product, or category is irreversible. All associated cloud resources will be destroyed by OpenTofu. Verify there are no active workloads before proceeding.

Chapter 16

Introducing a New Product

This chapter describes the complete process for making a new infrastructure product available in the webshop. Follow the steps sequentially.

16.1 Overview

1. Plan the product
2. Prepare the OpenTofu module
3. Configure the GitLab webhook
4. Create the product in the Admin UI
5. Define parameters (import or manual)
6. Configure deployment environments and pricing
7. Set up AI translations
8. Test the full order flow
9. Publish

16.2 Step 1: Plan the Product

Before touching any UI, answer these questions:

- **What does the product deploy?** (e.g. *Managed PostgreSQL VM on Linode*)
- **Which OpenTofu module delivers it?** (`vm-platform`, `k8s-platform`, or a new module)
- **In which environments is it available?** (e.g. Linode Frankfurt, Linode Amsterdam)
- **What parameters does the user need to provide?** (e.g. `vm_label`, `instance_type`, `db_name`)

- What is the base price in EUR per environment?
- Who assigns the cost center? (project / select / overhead)

16.3 Step 2: Prepare the OpenTofu Module

If the product uses an existing module (`vm-platform` or `k8s-platform`), skip to Step 3. For a new product type:

1. Create a new GitLab repository in the `open-hybrid-cloud` group.
2. Add the standard files: `provider.tf`, `backend.tf`, `variables.tf`, `main.tf`, `outputs.tf`, `.gitlab-ci.yml`.
3. Declare all user-provided parameters as `variable` blocks in `variables.tf` with `description`, `type`, and `default` where applicable:

```
variable "db_name" {
  description = "Name of the PostgreSQL database to create."
  type        = string
}

variable "db_size_gb" {
  description = "Database volume size in GB."
  type        = number
  default     = 20
}
```

4. Test the module locally:

```
tofu init -backend-config="address=..." \
          -backend-config="username=..." \
          -backend-config="password=..."
tofu plan -var="linode_token=$TOKEN" \
          -var="db_name=test" \
          -var="db_size_gb=20"
tofu apply
tofu destroy
```

5. Commit and push to the `main` branch.

16.4 Step 3: Configure the GitLab Webhook

In the GitLab repository:

1. Go to **Settings** → **CI/CD** → **Pipeline trigger tokens**.
2. Click **Add new token** and give it a descriptive name (e.g. *webshop-prod*).
3. Note the **trigger URL** and **token**.

The trigger URL has the form:

```
https://gitlab.example.com/api/v4/projects/<PROJECT_ID>/ref/main/trigger/pipeline
```

16.5 Step 4: Create a Category (if needed)

Under **Administration** → **Categories**:

1. Click **New Category**.
2. Enter a **Name**.
3. Optionally assign a **Category Parameter Set** (see Step 5).
4. Save.

16.6 Step 5: Create the Product

Under **Administration** → **Products** → **New**:

16.6.1 Basic Information

1. Select the **Category**.
2. Enter the product **Name** in the primary language.
3. Enter a detailed **Description** in the primary language.
4. Upload a product **Image** (PNG or JPEG, max 10 MB).
5. Click **Save & Continue**.

16.6.2 Parameters

Choose one of two import methods:

Option A — Import from `variables.tf` (recommended):

1. Click **Browse Repository**.
2. Select the **GitLab Source** and the repository.
3. Select the branch (`main`) and navigate to `variables.tf`.
4. Click **Import Parameters**.
5. The webshop parses the HCL and creates parameter records for each variable. Review and adjust:
 - **Required** — must the user fill this in?
 - **Sensitive** — should the value be masked in the UI and audit log?
 - **Default value** — pre-fill the order form.

- **Type** — text, number, boolean, or dropdown.

Option B — Manual entry:

Click **Add Parameter** and fill in each field manually.

Note

Parameters whose names match OpenTofu variable names (e.g. `instance_type`) are passed to the webhook as `TF_VAR_instance_type`. The name in the webshop must match the `variable` block name in `variables.tf` exactly.

16.6.3 Deployment Environments

1. Click **Add Environment**.
2. Select an environment from the dropdown (must be pre-configured under Administration → Environments).
3. Enter the **Price** in the base currency.
4. For the webhook, either:
 - Use the environment's default webhook URL, or
 - Enter a product-specific webhook URL and token.
5. Configure the **Cost Center Mode**:
 - Project** Cost center inherited from the project.
 - Select** Orderer selects from the cost center list.
 - Overhead** Always assigns a fixed cost center.
6. Repeat for each environment.

16.6.4 Multiple Webhooks (Pipeline Arrays)

A single product-environment combination can have multiple webhooks executed in order (e.g. first provision the VM, then configure DNS, then register in Vault). Under the webhook section:

1. Click **Add Webhook**.
2. Enter URL, token, and **Execution Order** (1, 2, 3 ...).
3. Webhooks are triggered sequentially in ascending order.

16.7 Step 6: Configure AI Translation

1. Ensure an AI provider is configured (Chapter 7).
2. Open the product in **Administration** → **Products**.

3. Click **Generate AI Translation**.
4. Review the generated translations for all 25 languages.
5. Edit any that require adjustment.
6. Save.

Tip

Generate translations early in the process. The AI translates both the product name and description. Individual translations can be re-generated for specific languages by editing the translation tab directly.

16.8 Step 7: Global and Category Parameter Sets

Global parameters (under **Administration** → **Global Parameters**) apply to every order regardless of product. Use them for organization-wide fields such as a cost center code or project tag that must always be captured.

Category parameters (under **Administration** → **Categories** → **{category}** → **Parameters**) apply to all products within a category. Use them to avoid repeating parameters shared by related products (e.g. all VM products share a **region** parameter).

Parameter hierarchy (applied in this order on the order form):

1. Global parameters
2. Category parameters
3. Product parameters
4. Environment-specific parameters (override per-environment)

16.9 Step 8: Test the Full Order Flow

1. Log in as a **Admin** (Admins trigger webhooks directly).
2. Navigate to **Catalog** and find the new product.
3. Click **Configure**, fill in test parameters, and click **Deploy Now**.
4. Monitor the order under **My Orders** — the status should progress from *Provisioning* to *Completed*.
5. Verify the infrastructure was created in the Linode Cloud Manager.
6. Test the email notification (check the admin email inbox).
7. Test the **Decommission** flow:
 - Open **Infrastructure**.
 - Click **Decommission** on the test resource.

- Verify the destroy pipeline runs and the resource disappears from the Linode Cloud Manager.

Important

Always test decommission in a non-production environment first. A destroy pipeline permanently deletes cloud resources and their data.

16.10 Step 9: Test with a Project Manager Account

1. Log in (or create a test account) as a **Project Manager**.
2. Place an order for the new product.
3. Verify the order shows *Pending Approval*.
4. Switch to the **Admin** account.
5. Approve the order under **Approvals**.
6. Verify the webhook fires and provisioning completes.
7. Verify the Project Manager receives the confirmation email.

16.11 Step 10: Product is Live

Once testing passes:

- The product is immediately visible in the catalog.
- No deployment or restart of the webshop is required.
- Announce the new product to users via the notification service or the shop's subtitle/branding.

Chapter 17

Project Management

17.1 Projects

A *project* is a container for related infrastructure orders. Every order must be associated with a project. Projects have:

- A **Name** and optional **Description**.
- An optional **Cost Center** (used when cost center mode is *Project*).
- An owner (the user who created it).

Project Managers see only their own projects. **Admins** and **Root users** see all projects.

17.2 Infrastructure Overview

The **Infrastructure** page shows all deployed resources grouped by project and environment. Each resource shows:

- Product name and environment.
- Deployment date and current status.
- Key parameters (e.g. VM label, cluster name).
- Buttons: **Use as Template**, **Decommission**.

17.3 Cost Center Assignment

Cost centers are maintained under **Administration** → **Cost Centers**. The three assignment modes give fine-grained control over who can allocate costs:

Project No user input; cost center comes from the project. Use for products where cost is always allocated to the project budget.

Select Orderer picks from the cost center list. Use when individual orders can span cost centers.

Overhead Always assigns a fixed cost center regardless of order. Use for shared services (monitoring, DNS).

Appendix A

Supported Languages

The following 25 languages are supported in the UI and for AI product translation:

Code	Language	Native name
de	German	Deutsch
en	English	English
fr	French	Français
it	Italian	Italiano
es	Spanish	Español
pt	Portuguese	Português
nl	Dutch	Nederlands
pl	Polish	Polski
cs	Czech	Čeština
sk	Slovak	Slovenčina
hu	Hungarian	Magyar
ro	Romanian	Română
bg	Bulgarian	Balgarski (Cyrillic)
hr	Croatian	Hrvatski
sl	Slovenian	Slovenščina
et	Estonian	Eesti
lv	Latvian	Latviešu
lt	Lithuanian	Lietuvių
fi	Finnish	Suomi
sv	Swedish	Svenska
da	Danish	Dansk
el	Greek	Ellinika (Greek script)
mt	Maltese	Malti
ru	Russian	Russkij (Cyrillic)
uk	Ukrainian	Ukrayinska (Cyrillic)

Table A.1: Supported language codes

Appendix B

Environment Variable Reference

Variable	Description	Required
<code>DATABASE_URL</code>	PostgreSQL connection string (DSN)	Yes
<code>SESSION_KEY</code>	32-byte hex key for session encryption	Yes
<code>ADMIN_EMAIL</code>	Email of the initial Root account	Yes
<code>ADMIN_PASSWORD</code>	Password of the initial admin account	Yes
<code>BASE_URL</code>	Public base URL (e.g. <code>https://shop.example.com</code>)	Yes
<code>PORT</code>	HTTP listen port (default: 8080)	No
<code>OIDC_CLIENT_ID</code>	Entra ID application (client) ID	No
<code>OIDC_CLIENT_SECRET</code>	Entra ID client secret	No
<code>OIDC_ISSUER</code>	OIDC issuer URL (<code>https://login.microsoft...</code>)	No
<code>SMTP_HOST</code>	SMTP server hostname	No
<code>SMTP_PORT</code>	SMTP port (default: 587)	No
<code>SMTP_FROM</code>	Sender email address	No
<code>SMTP_USERNAME</code>	SMTP authentication username	No
<code>SMTP_PASSWORD</code>	SMTP authentication password	No
<code>SMTP_TLS</code>	<code>true</code> to use TLS/SSL	No
<code>AI_PROVIDER</code>	AI translation provider at startup	No
<code>AI_ENDPOINT</code>	AI API endpoint URL	No
<code>AI_API_KEY</code>	AI API key	No
<code>AI_MODEL</code>	AI model name	No
<code>EXCHANGE_RATE_KEY</code>	Exchange rate API key	No

Table B.1: Environment variable reference

Appendix C

OpenTofu Module Variable Reference

C.1 network-core Variables

Variable	Type	Description	Default
linode_token	string	Linode API token (sensitive)	—
region	string	Linode region	eu-central
vpc_label	string	VPC label	ohc-vpc-prod
vpc_description	string	VPC description	(see defaults)
subnets	map(obj)	Subnet map (label→cidr)	app + mgmt
firewall_inbound_rules	list(obj)	Inbound firewall rules	SSH + ICMP
dns_domain	string	DNS zone domain (empty=none)	""
tags	list	Resource tags	["infra", "network-core"]

Table C.1: network-core variables

C.2 vm-platform Variables

Variable	Type	Description	Default
linode_token	string	Linode API token (sensitive)	—
vm_label	string	Unique VM label (state name)	—
region	string	Linode region	eu-central
vpc_id	number	VPC ID from network-core	—
subnet_id	number	Subnet ID from network-core	—
firewall_id	number	Base firewall ID	—
instance_type	string	Linode plan	g6-standard-2
image	string	Boot image	linode/debian12
root_pass	string	Root password (sensitive)	—
ssh_keys	list	Authorized SSH keys	[]
volume_size_gb	number	Block storage size (0=none)	0
additional_inbound_rules	list(obj)	Extra firewall rules	[]
dns_zone_id	number	DNS zone ID (null=none)	null
dns_hostname	string	A record hostname	""

Variable	Type	Description	Default
tags	list	Resource tags	["infra", "vm"]

Table C.2: vm-platform variables

C.3 k8s-platform Variables

Variable	Type	Description	Default
linode_token	string	Linode API token (sensitive)	—
cluster_label	string	Unique cluster label	—
region	string	Linode region	eu-central
kubernetes_version	string	K8s version	1.31
vpc_id	number	VPC ID	—
subnet_id	number	Subnet ID	—
node_pools	list(obj)	Node pool configurations	1× g6-standard-2
additional_inbound_rules	list(obj)	Extra firewall rules	[]
dns_zone_id	number	DNS zone ID (null=none)	null
dns_hostname	string	A record hostname	""
tags	list	Resource tags	["infra", "k8s"]

Table C.3: k8s-platform variables

C.4 shared-services Variables

Variable	Type	Description	Default
linode_token	string	Linode API token (sensitive)	—
region	string	Linode region	eu-central
gitlab_label	string	GitLab CE Linode label	ohc-gitlab
gitlab_instance_type	string	Linode plan for GitLab CE	g6-standard-4
vault_label	string	Vault Linode label	ohc-vault
vault_instance_type	string	Linode plan for Vault	g6-standard-2
image	string	Boot image	linode/debian12
root_pass	string	Root password (sensitive)	—
ssh_keys	list	Authorized SSH keys	[]
vpc_id	number	VPC ID from network-core	—
subnet_id	number	Subnet ID from network-core	—
allowed_admin_ipv4	list	IPv4 CIDRs for SSH & Vault admin	["0.0.0.0/0"]
tags	list	Resource tags	["infra", "shared"]

Table C.4: shared-services variables

C.5 observability Variables

Variable	Type	Description	Default
linode_token	string	Linode API token (sensitive)	—
region	string	Linode region	eu-central
obs_label	string	Linode instance label	ohc-obs
instance_type	string	Linode plan	g6-standard-2
image	string	Boot image	linode/debian12
root_pass	string	Root password (sensitive)	—
ssh_keys	list	Authorized SSH keys	[]
vpc_id	number	VPC ID from network-core	—
subnet_id	number	Subnet ID from network-core	—
volume_size_gb	number	Block storage size in GiB	200
grafana_port	number	Grafana HTTP port	3000
allowed_grafana_ipv4	list	IPv4 CIDRs for Grafana access	["0.0.0.0/0"]
tags	list	Resource tags	["infra","obs"]

Table C.5: observability variables

Appendix D

HTTP Route Reference

All routes are registered in `src/internal/handler/routes.go`. The **Role** column uses the following abbreviations:

- **public** — no authentication required
- **auth** — any authenticated user
- **admin** — role `admin` or higher
- **root** — role `root` only

Method	Path	Description	Role
<i>Authentication</i>			
GET	/login	Login page	public
POST	/login	Submit credentials	public
GET	/auth/callback	OIDC / Entra ID callback	public
POST	/logout	Destroy session	auth
POST	/lang	Switch UI language	auth
<i>General (any authenticated user)</i>			
GET	/	Home / dashboard	auth
GET	/catalog	Product catalog	auth
GET	/catalog/{id}	Product detail page	auth
GET	/products/{id}/image	Serve product image	auth
GET	/branding/logo	Serve shop logo	auth
GET	/impressum	Imprint page	auth
GET	/settings/profile	Profile & password change	auth
POST	/settings/profile	Save profile / new password	auth
<i>Orders</i>			
GET	/orders	My orders list	auth
GET	/orders/new	New order form	auth
POST	/orders/new	Submit order	auth
GET	/orders/{id}	Order detail / status	auth
<i>Projects</i>			
GET	/projects	Projects list	auth
GET	/projects/new	New project form	auth
POST	/projects/new	Create project	auth
GET	/projects/{id}	Project detail	auth

Method	Path	Description	Role
POST	/projects/{id}	Update project	auth
POST	/projects/{id}/delete	Delete project	auth
<i>Infrastructure</i>			
GET	/infrastructure	Infrastructure overview	auth
POST	/infrastructure/{id}/decommission	Trigger decommissioning	auth
<i>Approvals & Audit (role: admin)</i>			
GET	/approvals	Approval queue	admin
POST	/approvals/{id}/approve	Approve order	admin
POST	/approvals/{id}/reject	Reject order with note	admin
GET	/audit	Audit log viewer	admin
GET	/audit/export	Export audit log (PDF)	admin
<i>Admin — Categories (role: shop_admin)</i>			
GET	/admin	Admin dashboard	shop_admin
GET	/admin/categories	Category list	shop_admin
POST	/admin/categories	Create category	shop_admin
POST	/admin/categories/{id}/delete	Delete category	shop_admin
GET	/admin/categories/{id}/parameters	Category-level parameters	shop_admin
POST	/admin/categories/{id}/parameters	Add category parameter	shop_admin
POST	/admin/categories/{id}/parameters/{pid}/delete	Delete category parameter	shop_admin
<i>Admin — Products (role: shop_admin)</i>			
GET	/admin/products	Product list	shop_admin
GET	/admin/products/new	New product form	shop_admin
POST	/admin/products/new	Create product	shop_admin
GET	/admin/products/{id}	Edit product	shop_admin
POST	/admin/products/{id}	Update product	shop_admin
POST	/admin/products/{id}/delete	Delete product	shop_admin
POST	/admin/products/{id}/translate	Trigger AI translation	shop_admin
GET	/admin/products/{id}/image	Image upload page	shop_admin
POST	/admin/products/{id}/image	Upload product image	shop_admin
POST	/admin/products/{id}/environments	Link product to environment	shop_admin
POST	/admin/products/{id}/parameters	Add product parameter	shop_admin
POST	/admin/products/{id}/parameters/delete-all	Delete all product parameters	shop_admin
POST	/admin/products/{id}/parameters/{pid}/delete	Delete single product parameter	shop_admin
POST	/admin/products/{id}/webhooks	Add product webhook	shop_admin
POST	/admin/products/{id}/webhooks/{wid}/delete	Delete product webhook	shop_admin
<i>Admin — Environments, Sources, Cost Centers (role: shop_admin)</i>			
GET	/admin/environments	Deployment environment list	shop_admin
POST	/admin/environments	Create environment	shop_admin
POST	/admin/environments/{id}/delete	Delete environment	shop_admin
GET	/admin/sources	GitLab sources list	shop_admin
POST	/admin/sources	Create GitLab source	shop_admin
POST	/admin/sources/{id}/delete	Delete GitLab source	shop_admin
GET	/admin/costcenters	Cost center list	shop_admin
POST	/admin/costcenters	Create cost center	shop_admin
POST	/admin/costcenters/{id}/delete	Delete cost center	shop_admin
<i>Admin — Users (role: shop_admin)</i>			
GET	/admin/users	User list	shop_admin
POST	/admin/users	Create user	shop_admin
GET	/admin/users/{id}/edit	Edit user form	shop_admin

Method	Path	Description	Role
POST	/admin/users/{id}/edit	Save user changes	shop_admin
POST	/admin/users/{id}/deactivate	Deactivate user	shop_admin
POST	/admin/users/{id}/delete	Delete user	shop_admin
<i>Admin — Parameters, Currencies, GitLab (role: shop_admin)</i>			
GET	/admin/parameters	Global parameter list	shop_admin
POST	/admin/parameters	Create global parameter	shop_admin
POST	/admin/parameters/{id}/delete	Delete global parameter	shop_admin
GET	/admin/currencies	Exchange rate management	shop_admin
POST	/admin/currencies/refresh	Refresh exchange rates	shop_admin
GET	/admin/gitlab/projects	Browse GitLab projects	shop_admin
GET	/admin/gitlab/branches	(AJAX)	shop_admin
		Browse GitLab branches	
GET	/admin/gitlab/files	Browse GitLab files (AJAX)	shop_admin
POST	/admin/gitlab/import-vars	Import variables from .tf file	shop_admin
<i>Admin — Configuration (role: shop_admin)</i>			
GET	/admin/smtp	SMTP configuration	shop_admin
POST	/admin/smtp	Save SMTP settings	shop_admin
GET	/admin/ai-config	AI translation configuration	shop_admin
POST	/admin/ai-config	Save AI configuration	shop_admin
GET	/admin/branding	Shop branding	shop_admin
POST	/admin/branding	Save branding settings	shop_admin

Table D.1: Complete HTTP route reference

Appendix E

Requirements Traceability Matrix

The following table maps each functional and non-functional requirement to the implementation location and the handbook section that describes it.

ID	Requirement	Implementation	Section
FA-01	User authentication (local + SSO)	internal/service/impl/user.go , internal/handler/auth.go	§8
FA-02	Role-based access control (admin, user, pm)	internal/middleware/auth.go , internal/model/user.go	§8
FA-03	Product catalog with categories	internal/handler/catalog.go	§15
FA-04	Configurable product parameters	internal/model/parameter.go , admin UI	§16.6
FA-05	Shopping-cart order flow	internal/handler/order.go	§15
FA-06	Order approval workflow	internal/handler/approval.go	§15
FA-07	GitLab webhook trigger	internal/service/impl/pipeline.go	§4.5
FA-08	Pipeline status polling	internal/polling/ , internal/service/impl/pipeline.go	§4.5
FA-09	Infrastructure overview	internal/handler/infrastructure.go	§17.2
FA-10	Decommission (destroy) flow	internal/handler/infrastructure.go	§15.5
FA-09.5–9.8	Cascade decommission on project/product/category delete	internal/service/impl/infrastructure.go	§15.5
FA-11	Audit log	internal/audit/ , internal/handler/audit.go	§9
FA-12	AI-powered product translation	internal/service/impl/translation.go	§7.4
FA-13	Multi-currency support	internal/service/impl/currency.go	§7.6
FA-14	Branding configuration	internal/handler/admin.go , internal/model/branding.go	§7.5
NFA-01.1	Docker deployment	Dockerfile , infra/docker-host/	§6.3
NFA-01.2	Kubernetes deployment	infra/helm/open-hybrid-cloud/ manifests	§6.4
NFA-01.3	Single binary (embed.FS)	cmd/server/main.go , <code>//go:embed</code> directives	§3
NFA-01.4	Database migrations at startup	cmd/migrate/main.go , internal/migrations/	§11.6
NFA-02.1	HTTPS enforcement	Nginx config (reverse proxy, TLS termination)	§6.3
NFA-02.2	Secure session cookies	internal/session/ , AES-GCM encrypted <code>HttpOnly</code> cookies	§3
NFA-02.3	bcrypt password hashing	internal/service/impl/user.go	§3
NFA-02.4	Role-enforced routes	internal/middleware/auth.go	§8
NFA-03	25-language i18n	internal/i18n/i18n.go	§A
NFA-04	Audit trail for all state changes	internal/audit/service.go	§9

Table E.1: Requirements traceability matrix

Appendix F

Compiling This Handbook

The handbook source is at [docs/handbook.tex](#). Compile with:

```
# From the repo root
make docs

# Or manually (run twice for ToC/references)
cd docs
pdflatex handbook.tex
pdflatex handbook.tex
```

Required LaTeX packages: lmodern, geometry, fancyhdr, xcolor, tikz, pgf, booktabs, longtable, tabularx, listings, tcolorbox, hyperref, enumitem, float, caption, graphicx, microtype, parskip, pdfscape, pifont.

All packages are available in TeX Live 2023+ and MiKTeX 23+.